

Anaesthesia

World of Revision

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Instructions

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PRE - ANAESTHESIA CHECKUP

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PAC :

1. Detailed history.
2. Physical examination.
3. Complete systemic examination.
4. Airway examination.
5. Grade the patient.
6. Pre-operative orders and consent.

Co-Morbidities & their Anaesthetic Implications

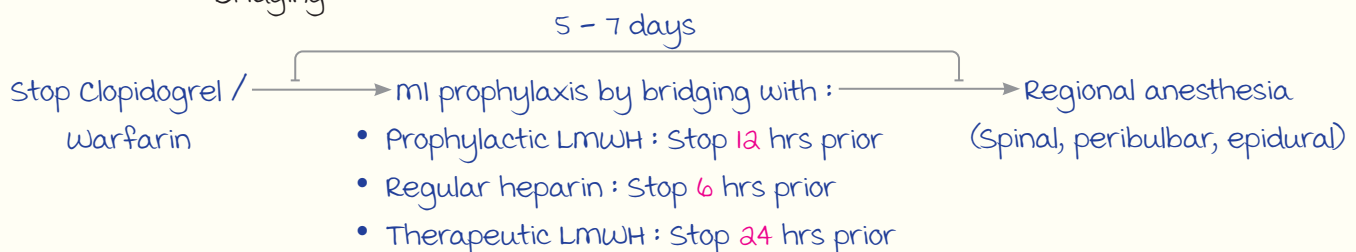
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Co-morbidities & drugs		Continued/Not
Hypertension	Antihypertensives	Continued
	ACEi, ARBs	Avoided in major surgeries (Severe hypotension)
Diabetes mellitus	- OHA - Insulin (Ultra short & short)	Stop on day of surgery (Hypoglycemia)
	SGLT-2 inhibitors	- Stop 72 hrs prior : Euglycemic ketoacidosis - Stop 24 hrs prior (minor sx)
	Insulin (Intermediate & long acting)	Reduce to 1/3 rd (if stopped : DKA/Hyperglycemic hyperosmolar coma)
	Intra-op : - Check GRBS : Every 30 mins - Hyperglycemia → Short acting insulin	
Thyroid disorder	Hypothyroid Rx	Continue all Rx (Prevents delayed recovery)
	Hyperthyroid Rx	Continue (Prevents thyroid storm risk)
	Note : Intra-op thyroid storm : → Stop Sx → Control HR : • Tachycardia • Hypertension • ↑ Temperature • Narrow QRS complex Esmolol ↓ Add Propranolol	
Epilepsy (D/t ↓ O _a / ↑ CO _a / Acidosis)	AEDs	- Continue - Check LFT
Herbal medications		Check LFT Abnormal → Repeat after 24 hrs
Anti-TB medications		Check LFT, continue all ATT
Psychiatric conditions	Psychiatric drugs	Continue

----- Active space -----

Co-morbidities & Drugs		Continued/Not
Psychiatric conditions	MAO inhibitors : Selegiline	- Interact with synthetic opioids : → Hypertensive crisis - meperidine (Pethidine) - Ephedrine - Tramadol - Stop older MAO ⊖ : 3 weeks
	Lithium (magnesium)	- Interact with muscle relaxants to ↑ duration mivacurium/Atracurium Yes → Continue Li ↓ No - Stop Li 24 - 48 hrs before
OCP	Estrogen pills : DVT	- High risk : → Stop 4 weeks before - Old age - Long bone fracture - Bed ridden - major Sx - Low risk or only on progesterone only pills → Continue
Steroid		Continue
Diuretics	Non-thiazides	Stop
	Thiazides	- Continue (Low potency) - Check serum electrolytes
- PAH - Erectile dysfunction	Sildenafil	- S/e : vasodilation → Severe hypotension - Stop 24 - 48 hrs prior
Coronary artery disease	All cardiac Rx	Continued
	Aspirin	- Continue if low dose + alone - Stop 3 days prior for major Sx
	Clopidogrel/ Warfarin	- Stop 5 - 7 days prior - Topical anaesthesia : Continue

Bridging :



Personal, Family and Allergy History

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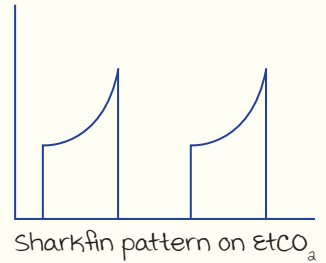
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Personal History :

Smoking :

- Stop 3 - 4 weeks prior.
- Complications :

	Intra-op bronchospasm	Intra-op laryngospasm
Presentation	Sudden unexplained : • Tachycardia • HTN • ↑ Airway resistance, wheeze ⊕ • Sharkfin pattern on EtCO_a	End of surgery : • Stridor • Rapid desaturation • Paradoxical chest wall movements • No air entry • Tachycardia → Bradycardia
Rx	Bronchodilators	• 100% O_a • AOC : Propofol • Severe : Succinylcholine



Alcohol :

Stop 24 - 48 hrs prior.

Tobacco chewing :

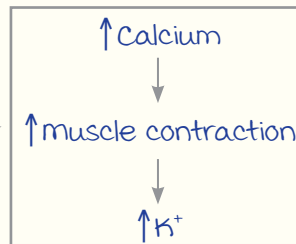
Submucosal fibrosis → **Restricted mouth opening** → Difficult intubation.

Family history : malignant Hyperthermia ❤️

Etiology :

Family h/o muscular dystrophy :
(mutation in **ryanodine receptor**)
+
inhalational agents/
succinylcholine

Pathophysiology :



c/f :

- Sudden unexplained tachycardia
- HTN
- ↑ Temperature
- EtCO_a (most sensitive) : **Step ladder pattern** ↑
- ECG : Wide QRS complex
- VF → Sudden cardiac arrest

mx :

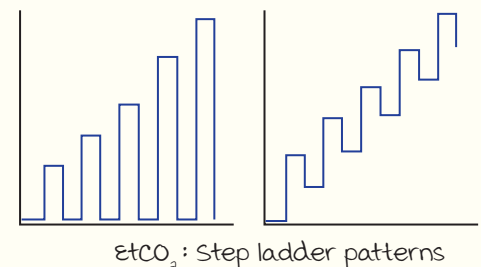
- 100% O_a.
- DOC : **Dantrolene sodium** 2.5 mg/kg diluted in distilled water.

mx of hyperkalemia :

- **Give calcium gluconate.**
- Add insulin + dextrose/ Salbutamol.

Complication :

Acute renal failure (myoglobin).



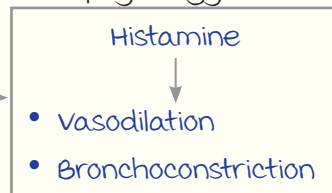
----- Active space -----

Allergy History : Anaphylactic Shock

Causes :

1. Antibiotic (m/c)
2. muscle relaxant
3. Chlorhexidine
4. Local anaesthetics

Pathophysiology :



c/f :

- Sudden unexplained tachycardia
- Hypotension
- ↑ Airway resistance, wheeze ⊕
- Redness, edema of face, airways

mx :

- DOC : Adrenaline (1 mL = 1 mg = 1 : 1000).
 - IV : 1 mL of 1 : 10,000.
 - IM : 0.5 mL of 1 : 1000.

Airway Examination

00:42:34

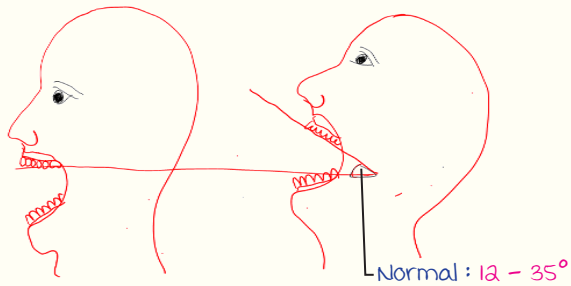
Mouth Opening :

Finger breadth techniques :

- 3 fingers : ⊕
- < 2 fingers : Chances of difficult intubation.

modified mallampati classification :

	Class I	Class II	Class III	Class IV
				
	I	II	III	IV
Hard palate	✓	✓	✓	✓
Soft palate	✓	✓	✓	✗
uvula	✓ Hanging freely	✓ Tip obscured	✗ Only base seen	✗
Fauces	✓	✗	✗	✗

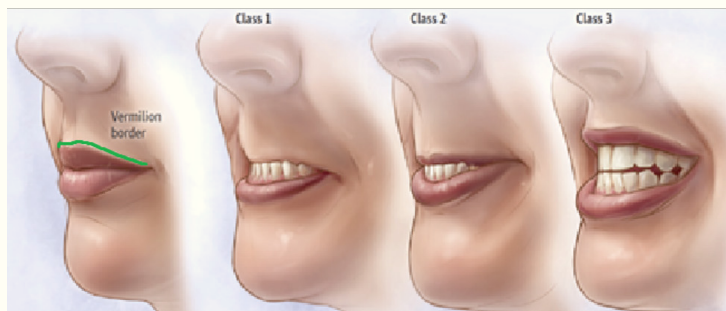
Neck mobility :

- Angle made by forehead from complete flexion-extension : $> 80^\circ$.
- Neck circumference : < 43 cm.



----- Active space -----

1. Thyromental distance : 6.5 cm
2. Sternomental distance : $12.5 - 13$ cm



upper lip bite test/mandibular protrusion test :

Cross vermillion border $\xrightarrow{\text{Yes}} \textcircled{N}$
 $\xrightarrow{\text{No}} \text{Receding mandible}$

American Society of Anaesthesiologists (ASA) Grading

00:51:02

	Criteria	Conditions
Grade I	Normal healthy patient	• Normal BMI • Non-smoker • Occasional alcoholic
Grade II	mild systemic disease (No functional limitation)	• mild-moderately obese • Smoker • Regular alcoholic • Well controlled disease • Pregnancy
Grade III	Severe systemic disease (with functional limitation)	• morbidly obese • Poorly controlled disease • Alcohol dependent • ESRD with regular dialysis • MI, CVA, stenting : > 3 months • On pacemaker

----- Active space -----

	Criteria	Conditions
Grade IV	Severe systemic disease : Life threatening	- Recent MI, CVA, stent : < 3 months - Unstable angina - Severe valvular disease - ESRD with no regular dialysis - ARDS - DIC
Grade V	moribund patient : Not expected to survive > 24 hrs	- Ruptured aneurysm - massive trauma - Ischemic bowel - MODS
Grade VI	Brain dead patient	-

Investigations

00:55:22

CBC :

Test	Procedure	minimum acceptable counts
Hb	Elective surgery	8 g/dL
	Elective with co-morbidity	10 g/dL
	Critically ill	12 g/dL
Platelet	Invasive procedure	50 K
	Central neuraxial block	1 lakh
	Peripheral neuraxial block	80 K

Other tests : RBS, serology viral markers, creatinine etc.

ECG and ECHO :

	ECHO	ECG
Not done	As routine investigation	- Asymptomatic patients - Low risk surgeries
may be considered	H/o LV dysfunction : Not evaluated for 1 year	- Asymptomatic pts w/o known CAD - Posted for intermediate-major procedures
Reasonable to perform	- SOB of unknown origin - Heart failure patients with worsening dyspnea	- K/c/o IHD - Significant arrhythmia, PAD, CVD - Significant structural heart disease

CXR :

- Only if LRTI + green/purulent sputum.
- Suspecting bullae.

CVS & RS Guidelines

01:00:18

----- Active space -----

Surgery post coronary stenting :

- Bare metal stent : Wait 30 days.
- Drug eluting stent : Wait 6 months.

Infective endocarditis prophylaxis :

- Past h/o IE.
- Prosthetic valves.
- unrepaired cyanotic congenital heart disease (CHD).
- Repaired CHD with residual defect.
- Cardiac transplant.

Cardiac Risk Stratification (ACC/AHA Guidelines) :

Emergency surgery : Consent → Sx.

Elective surgery :

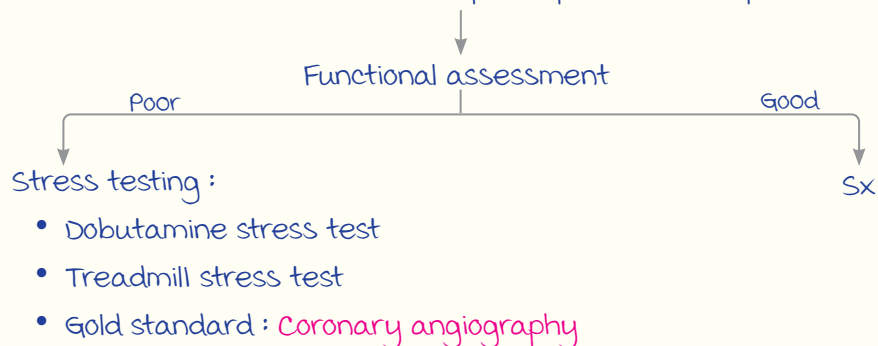
Postpone upto 60 days in :

- Acute coronary syndrome.
- Severe valvular heart disease.
- Decompensated heart failure.
- Significant arrhythmia.

Risk for MI/death :

1. Any major surgery (Intrapertioneal/suprainguinal/vascular/emergency).
2. H/o MI.
3. H/o CCF.
4. H/o CVA.
5. s. Creatinine > 2.
6. Diabetes on insulin.

Score > 3 → 12% risk of post-op cardiac complication.



----- Active space ----- Pulmonary Risk Stratification :

Risk factors :

Patient related :

- Old age
- ASA-PS class ≥ 2
- CHF
- Functionally dependent COPD
- Weight loss
- Impaired sensorium
- Cigarette smoker
- Alcohol use

Procedure related :

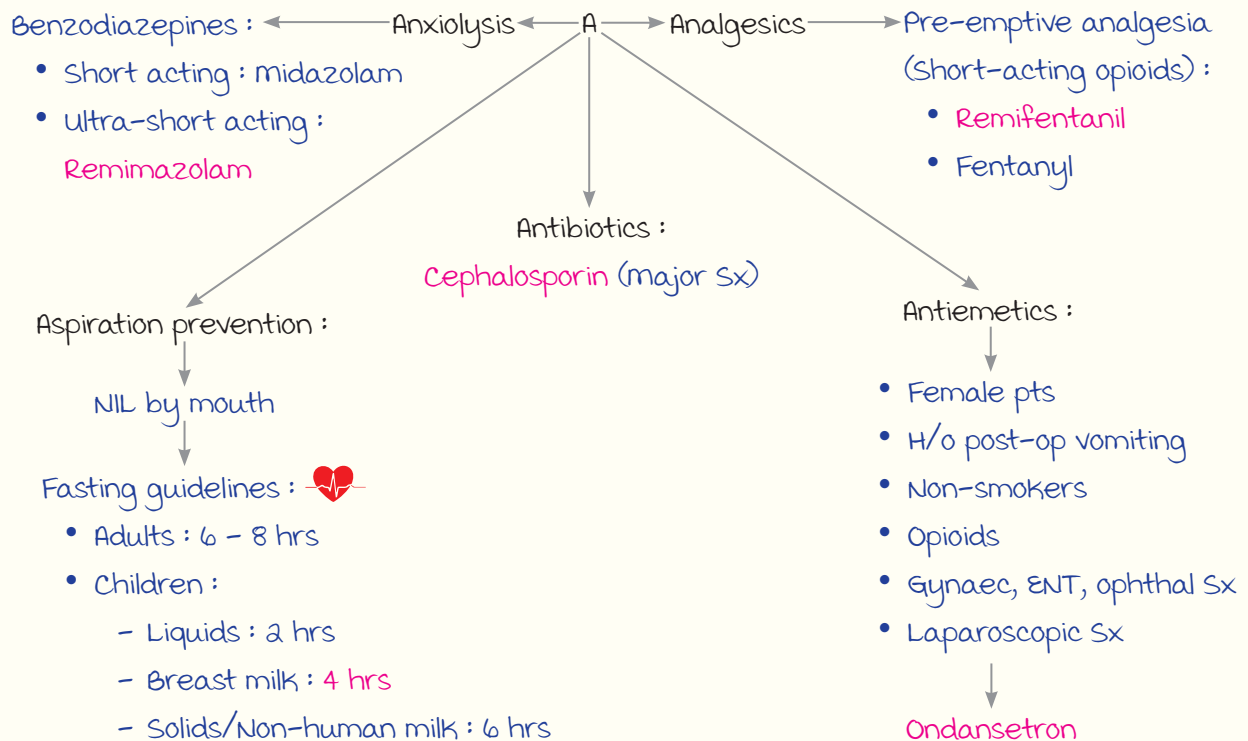
- Aortic aneurysm repair
- Thoracic Sx
- Abdominal, upper abdominal Sx
- Neuro Sx, head & neck Sx
- Emergency Sx
- Vascular Sx
- General anaesthesia
- Perioperative transfusion

Laboratory test :

- Albumin : < 3.5 g/dL
- CXR abnormalities
- BUN : > 7.5 mmol/L
(> 21 mg/dL)

Pre-Operative Medications

01:08:20



MONITORING UNDER GENERAL ANAESTHESIA

----- Active space -----

CNS Monitoring

00:00:40

Bispectral index :

- Depth of anaesthesia.
- Analyse EEG → Numerical value (0 - 100).



ECG

Cardiac Monitoring

00:03:16

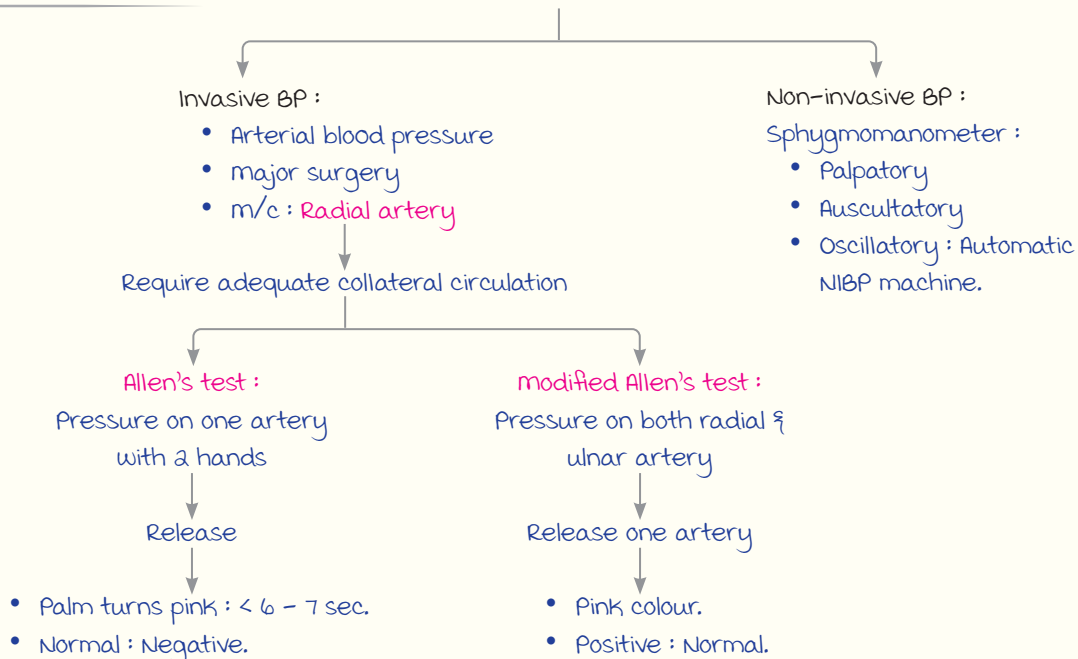
Heart Rate :

60 - 100 bpm.

ECG :

Lead	Importance	
II	Sensitive to arrhythmia	
V _P V _a	ST changes	Septal ischemia
V _S V ₆		Lateral wall ischemia

Blood Pressure :



----- Active space -----

Central Venous Pressure :

- Triple lumen catheter.
- **Cannot be** used for rapid resuscitation.

Use :

- Long term IV fluids.
- TPN.
- Dialysis.
- Inotrope/cardiac medications.

Dimension :

- Size : 7 Fr (21 G).
- Length : 20 cm.

Interpretation :

- $\textcircled{N}$ CVP : 0-5 cm H_2O ($\textcircled{N}$ Right heart function).
- $\uparrow$ CVP + $\downarrow$ BP : $\textcircled{R}$ Heart failure.
- $\downarrow$ CVP + $\downarrow$ BP : Fluids required.

Site :

- m/c : **Right IJV.**
- Femoral : $\uparrow$ Risk of infection (Least used).
- Left IJV :
 - Lung cupula injury : Pneumothorax.
 - Thoracic duct injury.

Technique : Seldinger technique.

Complication : Arrhythmia.

Pulmonary Capillary Wedge Pressure :**Swan-Ganz catheter** : IJV $\rightarrow$ RA $\rightarrow$ RV $\rightarrow$ PA $\rightarrow$ PC.

Use :

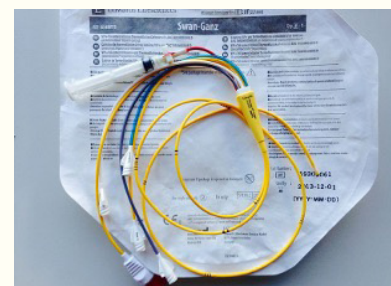
- Left heart function : 12 - 16 mmHg.
- Check core body temperature during cardiac Sx.

Complications :

- PA rupture.
- **Arrhythmia** m/c.



Triple lumen catheter



Swan-Ganz catheter

Respiratory System Monitoring

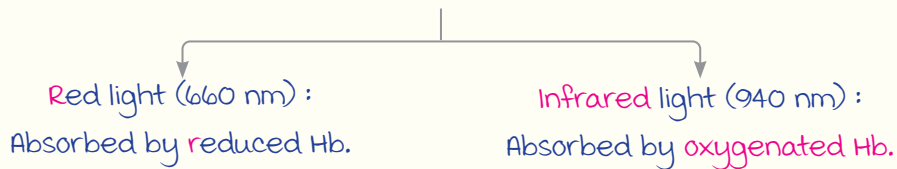
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Pulse Oximeter :

Amount of oxygen bound to Hb.

Principle (Beer Lambert's law) :



Normal : 98 - 100%.

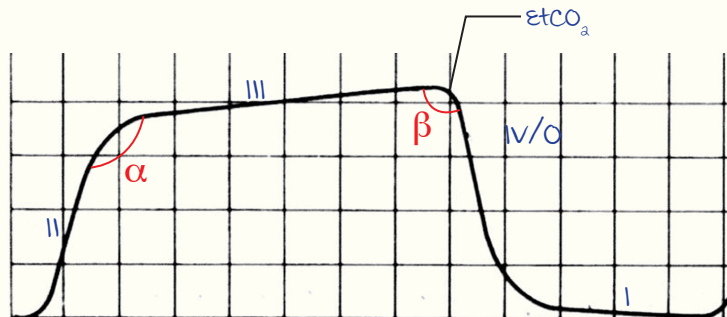
Limitations :

- Carboxy Hb : SpO_a falsely $\uparrow$ (Fire accidents).
- meth Hb : SpO_a falsely $\downarrow$, fixed at 85% (Iron in Fe^{3+} form).

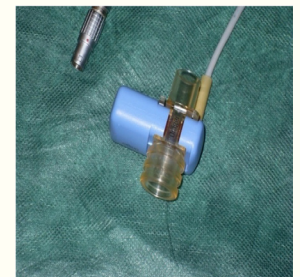
Capnography :

Exhaled CO_a $\rightarrow$ Graphical representation : Capnography.
 $\rightarrow$ measurement : Capnometry.

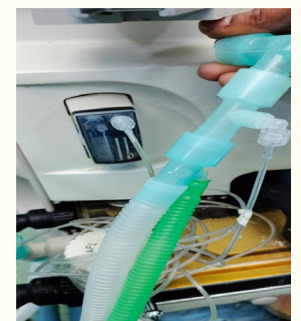
Normal :



Pulse oximeter



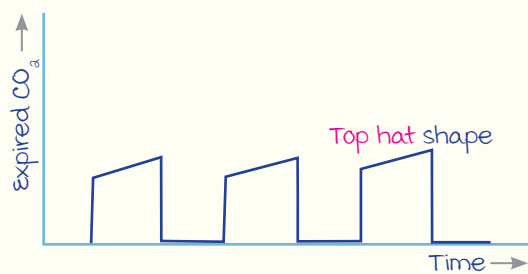
mainstream capnography



Side stream capnography

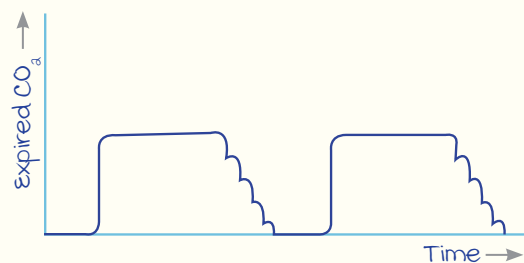
Phase	Capnograph	Physiology
Phase I	Inspiratory base line	• No gas exchange • No CO_a
Phase II	Expiratory upstroke	mixing of dead space gas with upper alveoli gas (CO_a rich)
Phase III	Alveolar plateau	• middle and lower alveoli CO_a is exhaled • Slightly positive upstroke
Phase IV/O	Inspiratory downstroke	Inspiration started

----- Active space ----- Capnographs :



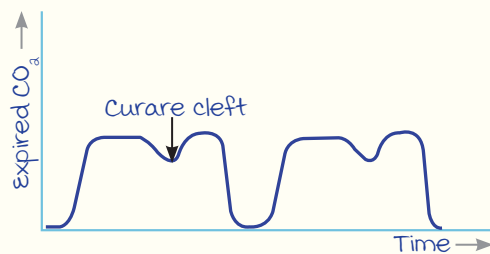
Normal :

- 35 - 45 mmHg.
- Tracheal intubation : ≥ 3 curves.

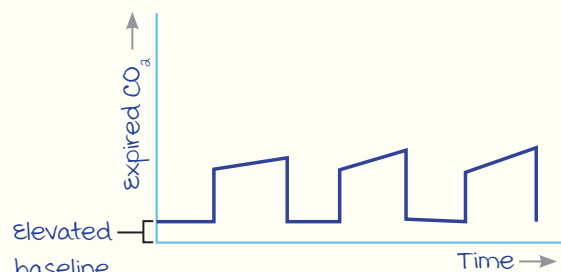


Cardiogenic oscillations :

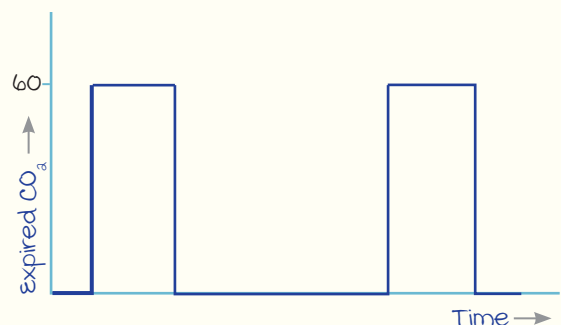
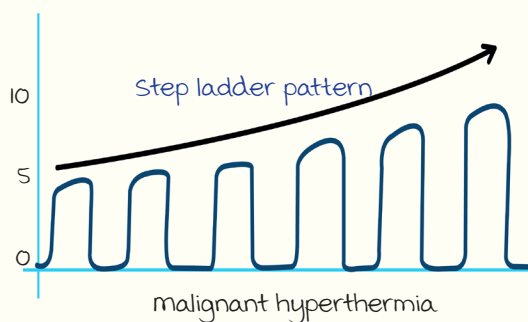
Physiological in children
(D/t thin chest wall).



Recovering from the effect
of muscle relaxant :
Spontaneous breathing efforts.

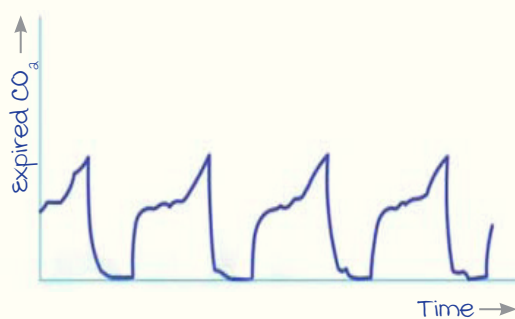


Rebreathing of CO_2 :
Exhausted soda lime/
inadequate fresh gas flow.

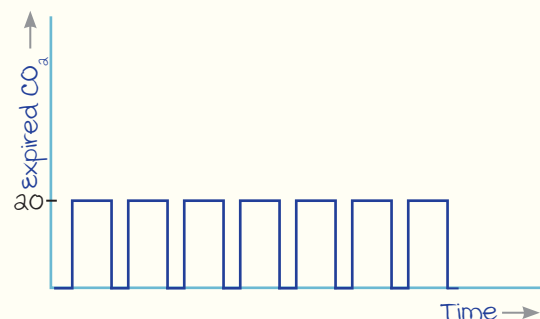


Hypoventilation :

- Opioid poisoning (CNS depressant)
- Baseline touching, $\uparrow \text{CO}_2$, $\downarrow \text{RR}$.

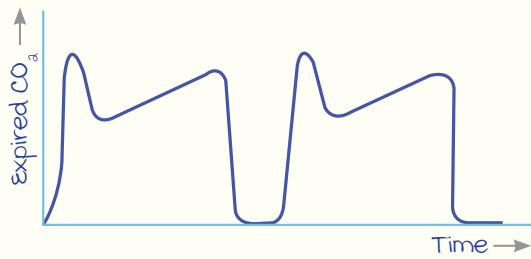


Dual plateau sign :
Leaky sample line.

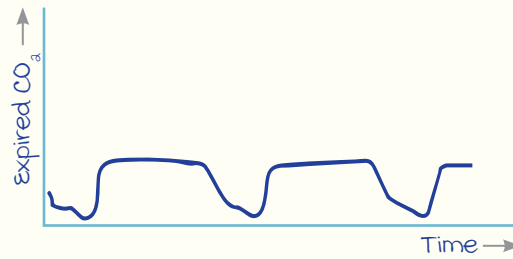


Hyperventilation :
Baseline touching, $\downarrow \text{CO}_2$, $\uparrow \text{RR}$.

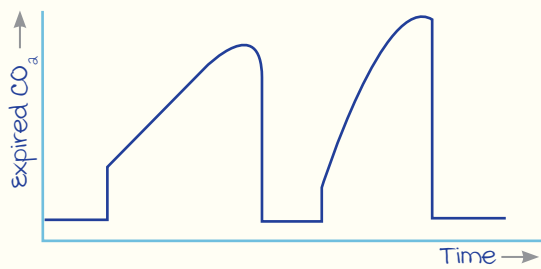
----- Active space -----



Single lung transplant :
2 peaks in phase III

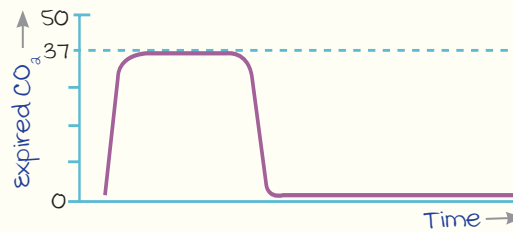


Incompetent inspiratory valve :
Splaying of phase IV.

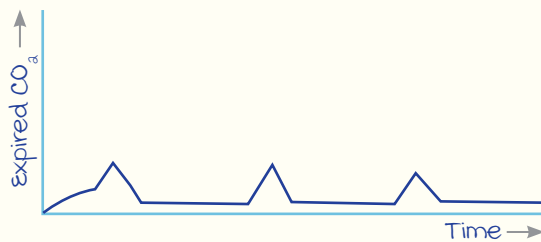


Bronchospasm/
partially obstructed ET tube :

- Increased upstroke of phase III.
- Shark fin pattern.



Sudden zeroing of EtCO₂ :
Accidental extubation/
circuit disconnection (m/c).



Intubation into esophagus

----- Active space -----

Neuromuscular Monitoring

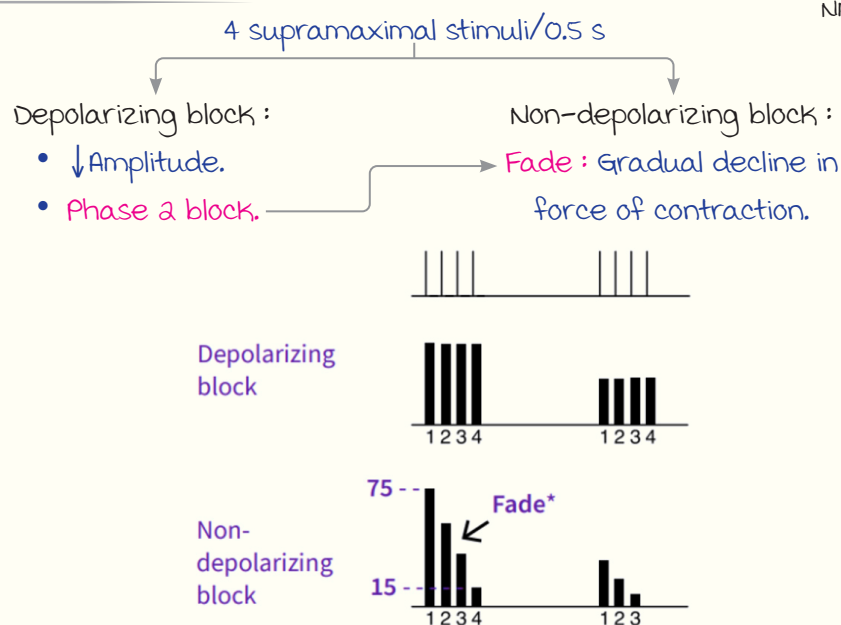
00:41:34

- Supramaximal stimuli.
- m/c nerve : **ulnar nerve**.
- m/c muscle : **Adductor pollicis**.



NM monitoring

Train of Four (ToF) :



ToF number > 0.9 : Fit for extubation.

Temperature Monitoring

00:46:54

- Hypothermia : Common.
- Hyperthermia (Rare) :
 - Thyroid storm.
 - malignant hyperthermia.

Core temperature :

- **Lower esophageal temperature** (m/c).
- Neurosurgery : Tympanic membrane/nasopharynx.
- Cardiac surgery : Pulmonary artery temperature (most accurate).

Rectal temperature :

- Intermediate b/w surface & core temperature
- Casualty, wards.

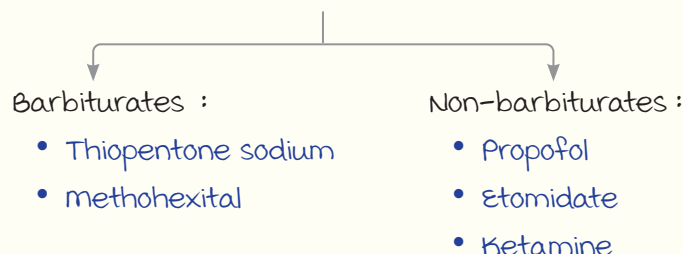
IV AND INHALATIONAL INDUCTION AGENTS

----- Active space -----

Receptor : **GABA**.

Action : Depressants ($\downarrow$ HR, $\downarrow$ BP, $\downarrow$ RR, $\downarrow$ liver & renal blood flow).

Classification :



Barbiturates

00:03:58

Thiopentone Sodium :

pH : 10.5 (Highly alkaline) $\rightarrow$ Do not mix with other drugs.

Dose : **3-5** mg/kg.

Features :

Mnemonic : **THIOPENTONE**.

	Features
T	Timing : • 15 sec • Arm-brain circulation time
H	Hypnosis (Low dose : Truth serum)
I	ICP reduction, anti-epileptic activity
O	Onion/garlic smell (D/t sulfur)
P	Potent GABA modulators : $\uparrow$ Duration of opening of channels
E	- Extravasation : Accidental intra-arterial injection (Cubital fossa) $\downarrow$ Gangrene - Rx : - Do not remove needle (Causes vasospasm) - Inject saline/heparin Stellate ganglion block (Lower cervical sympathetic ganglion) $\rightarrow$ Vasodilatation

	Features
N	Neurosurgery (AOC)
T	Anti-thyroid activity : AOC in hyperthyroidism
O	Older agent (Since 1934)
N	-
E	- End of effect by redistribution (High lipid solubility) - After 10-12 hrs $\downarrow$ metabolized by liver

Note :

Benzodiazepines : $\uparrow$ Frequency of opening of GABA channels.

----- Active space -----

methohexital :

Dose : 1 - 1.5 mg/kg (more potent than Thiopentone).

Indication : ECT

C/I : Neuro-Sx (Pro-epileptic d/t methyl group).

Barbiturate Poisoning :

- No antidote.
- Rx :
 - Supportive (Intubation).
 - Forced alkaline diuresis.

Propofol

00:17:25

Features :

- Antiemetic & antipruritic
- mnemonic : PROPOFOL.

	Features
P 	Preferred in : • Day care Sx. • Ambulatory Sx/Office based anaesthesia. • monitored anaesthesia care. • TIVA (Propofol + Remifentanyl).
R	Redistribution : • Lipid solubility. • Recovery time : 8 - 10 mins.
O	Onset : 15 seconds (Arm-brain circulation time)
P 	• Pleasant hallucinations (↑Addiction potential) • Painful injection : ↑ - Inject in larger veins - mixed with lignocaine
O	• Oily : Soyabean oil. • white : Egg lecithin (use < 6 hrs).
F	Faster recovery (No hangover)
O	Ocular Sx (↓IOP : max.)
L	Laryngospasm Rx : AOC (Blunts airway reflexes)

Propofol Infusion Syndrome :

Avoid propofol infusion ≥ 48 hrs.

C/f :

- Green colour urine.
- metabolic acidosis.
- Asystole.

Etomidate

00:25:42

----- Active space -----

Dose :

0.2 - 0.3 mg/kg.

Features :mnemonic : **ETOMIDate**.

	Features
E	• Emetic • Epileptogenic
T	Painful (Tortures the hand)
O	Oily (Propylene glycol)
m	myoclonus, minimal depressant
I	Inhibits adrenocortical synthesis
Date	• most cardiostable • AOC : Cardiac Sx, aneurysm Sx

Ketamine

00:28:50

MOA  : NMDA receptor antagonist (Stimulant) → ↑ Catecholamine levels.↓
Dissociative anaesthesia.**Features :**mnemonic : **KETAMINE**.

	Features
K	Kids : Im
E	Eyes : Nystagmus
T	• Thalamo-cortical junction : Site of action • Impulses from thalamus → Do not enter cortex ↓ Dissociation.
A 	Analgesic : • Only IV agent. • minor painful procedures.
m	• memory loss (Amnesia). • meals taken (No pre-op fasting).
I 	Increases : • HR, BP : Hypotensive situations • RR : Low resource setups • Salivation : Add anticholinergics (Atropine/glycopyrrolate) • ICP : Avoid in head injury & neuro-Sx • IOP : Avoid in ocular-Sx

----- Active space -----

	Features
N	NMDA receptor antagonist
ε 	- Excellent bronchodilator : AOC in Asthma & COPD - Emergence reactions : - unpleasant hallucinations. - Prevention : Add midazolam.

Recent Updates :

Approved for use in :

- Chronic depression.
- Opioid tolerance.

Routes of Administration & Dosages :

- IV : 1 - 2 mg/kg.
- Im : 4 - 6 mg/kg.
- Regional anaesthesia : mixed with LA (Prolongs action).

Inhalational Anesthetic Agents - General Characteristics

00:00:06

Purpose :

- Induce sleep in children.
- maintain depth of anaesthesia.

mode of delivery :

Inhalational agents are volatile liquids



Combine with gases in vapourizers



Vapours



Enter & exit through lungs.



Anaesthesia workstation

Classification :

Older agents (Flammable) :

- ether.
- Chloroform.

Newer agents (Non-flammable) :

- Halothane.
- Desflurane.
- Isoflurane.
- Sevoflurane.

Mechanism of Action :

Unknown.

Meyer-Overton rule :

Potency $\propto$ Lipid solubility

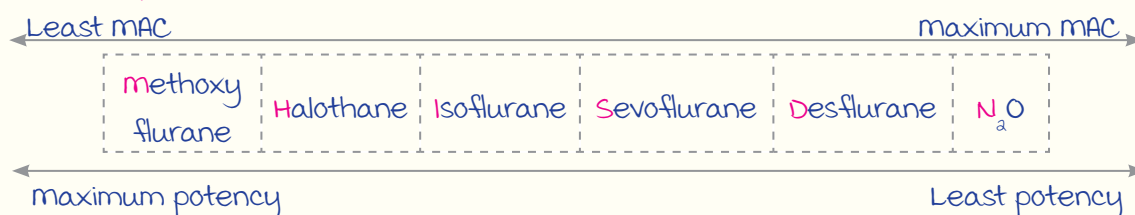
minimum alveolar concentration (MAC)  :

----- Active space -----

- **minimum** amount of agent required to produce immobility to painful stimulus in **50%** individuals.

$$\text{MAC} \propto \frac{1}{\text{Potency}}$$

- **Examples :**



Factors Affecting Onset & Recovery

00:48:27

During Onset :

Concentration effect :

↑ Inspired concentration → **Quicker** induction.

Second gas effect :

Inhalational agent + N₂O → ↑ Concentration of inhalational agent in lungs → **Quicker** induction.

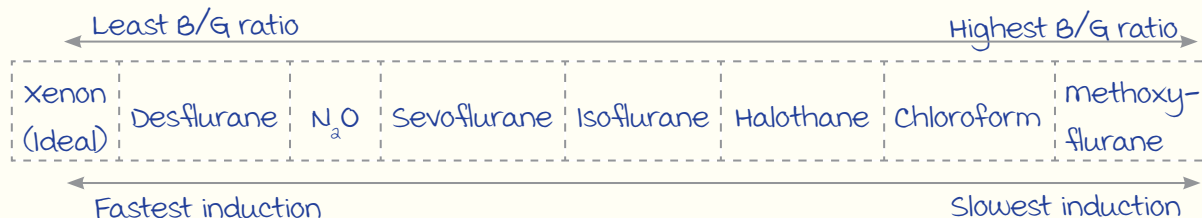
↓

 Rapid diffusion → ↑ uptake of inhalational agents
 capacity (Only N₂O) into pulmonary circulation

Blood gas partition coefficient :

$$\text{B/G ratio} = \frac{\text{Concentration of agent in blood}}{\text{Concentration of agent in alveoli}}$$

↑ B/G ratio → ↑ Concentration in blood → ↑ Solubility in blood → **Delayed** induction.



During Recovery :

Diffusion hypoxia (Fink effect) :

Rapid diffusion of N₂O from pulmonary circulation → ↑ Alveoli size → Dilution of O_a (↓ Concentration) → Hypoxia.

Rx : O₂ supplementation.

----- Active space -----

Systemic Effects

01:04:00

	Halothane	Isoflurane	Desflurane	Sevoflurane
RS	Bronchodilators			
	- max. bronchodilation - Sweet smelling : Children : 2nd AOC	- Bronchospasm - Laryngospasm	} more in desflurane - Irritant - Daycare Sx : 2nd AOC (D/t quick recovery)	Sweet smelling : - Asthmatics : AOC - Children : 1st AOC - Day Care Sx : 1st AOC
		Pungent odor		
CVS : ↓HR & ↓BP	- max. ↓ HR - max. arrhythmogenic with adrenaline	- AOC - Coronary steal phenomenon	Initial tachycardia ↓ Avoid	AOC (more preferred)
CNS	- Cerebral vasodilators → ↑ICP ↓ Cerebral metabolic O₂ consumption			
	max ↑ICP	Neurosurgery : ↑RR → ↓EtCO₂ (32 - 35 mmHg) → ↓ICP - Enflurane : Epileptogenic (Avoided)		
GIT, liver & biliary tract : ↓LBF	- max ↓LBF - max metabolism ↓ metabolites react with liver cells ↓ Halothane hepatitis : - Females - 40 yrs - Obese - multiple halothane exposure	Equally ↓LBF		
		-	Least metabolised	-
Renal system : ↓RBF	Fluoride → ↑ Nephrotoxicity (makes agents non-flammable) (max : methoxyflurane)			
	-	Preferred agents	- max. fluoride - unproven as nephrotoxic - Sevo + soda lime ↓ Prolonged use Compound A (Nephrotoxic in rats)	
Reproductive system	uterine relaxants → PPH			
Eyes	↓ IOP			

Trilene :

- Analgesia (Old agent).
- Rx of trigeminal neuralgia.

----- Active space -----

Nitrous Oxide (N_2O)

01:19:39

Cylinder : Blue.

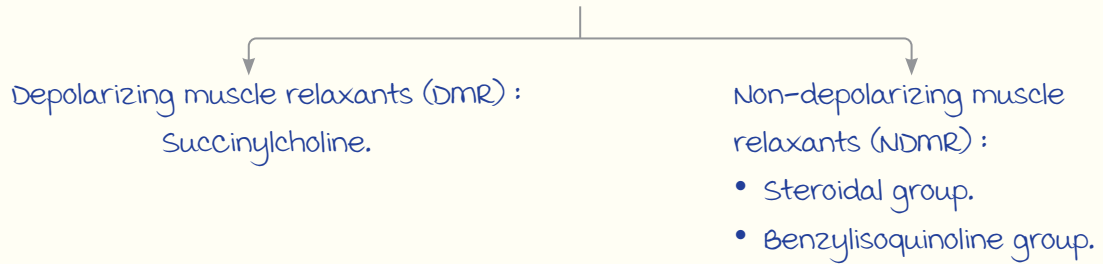
C/i :

- Interferes with vitamin B_{12} metabolism :
 - megaloblastic anemia.
 - Subacute combined degeneration of SC.
 - Pneumothorax
 - middle ear & retina sx
- } D/t rapid diffusion.

MUSCLE RELAXANTS

Indication : To facilitate airway management.

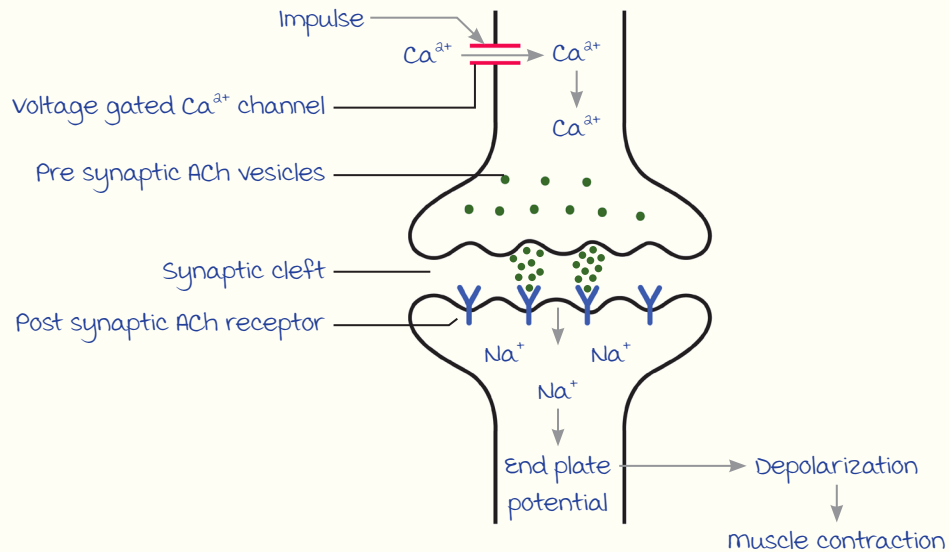
Types :



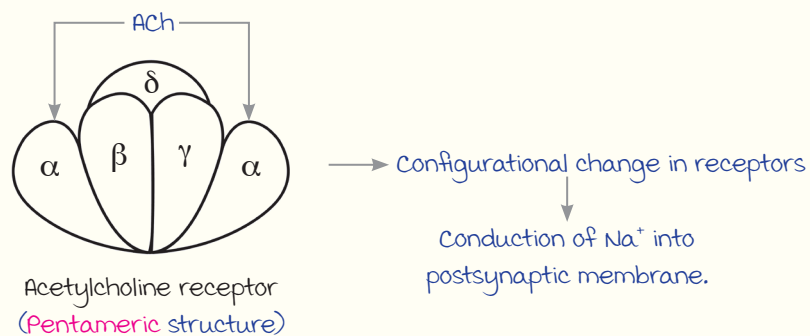
Neuromuscular Physiology

00:00:45

Events at NMJ :



Acetylcholine receptor :



Depolarizing Muscle Relaxants

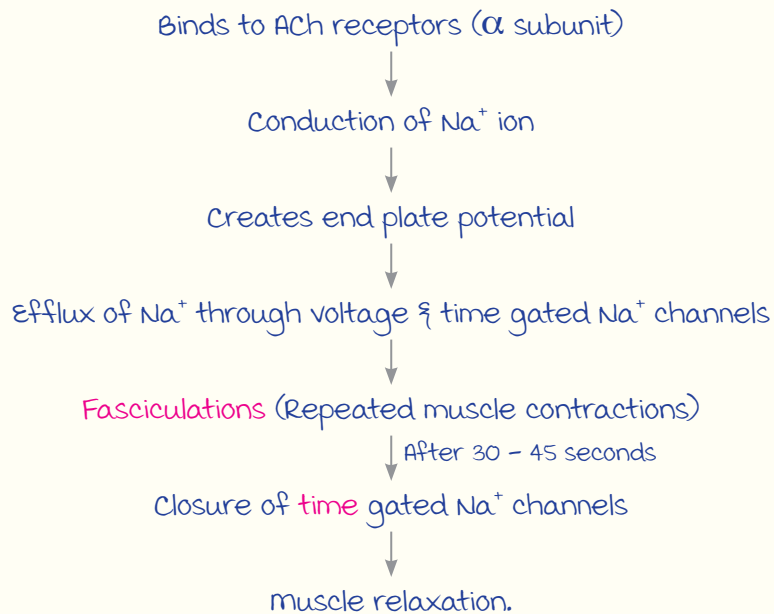
00:03:19

----- Active space -----

Succinylcholine :

Dose : 1 - 2 mg/kg.

MOA :



Features and clinical significance :

	Features	Clinical significance
Structure	Two molecules of ACh	Bradycardia ↓ Prevent with anti-cholinergics : - Atropine - Glycopyrrolate
Type of blockade	Non competitive	Cannot reverse with neostigmine
Onset of action	30 - 45 seconds	Rapid sequence intubation
Unique feature	muscle fasciculations	↑ ICP : Avoid in head injury and neurosurgery ↑ IOP : Avoid in eye surgeries ↑ Intra gastric pressure : Avoid in hiatus hernia - Post-op myalgia
metabolism	Pseudocholinesterase : - Production : Liver - Transport : Plasma	↓ Pseudocholinesterase ↓ Prolongation of action
Duration of action	< 10 mins	Anticipated difficult intubation

----- Active space -----

Prolonged action :

	Causes & mx	Duration of action
↓ Levels of pseudocholinesterase	• Liver diseases : Acute/chronic liver failure • Neonates, old age, pregnancy • Drugs : - Neostigmine - Pyridostigmine - Anti-cancer drugs	20 mins
Atypical pseudocholinesterase (Scoline apnea)	• ↓ metabolism of Sch • Genetically determined • Ix : Dibucaine number • Rx : - Continue mechanical ventilation - Fresh frozen plasma (FFP)	6 - 8 hrs
Phase II block (Resemble NDMR) ↓ Fade	• Large dose (≥ 5 mg/kg) → Receptor damage • Rx : - Continue mechanical ventilation - Wait for spontaneous recovery - Don't use neostigmine	-

Dibucaine number :

- Qualitative analysis of enzyme activity.
- Principle : Dibucaine (LA) $\xrightarrow{\text{Binds}}$ (N) Pseudocholinesterase.
- Homozygous atypical variant → 20 : 80.

c/i :

- Family h/o malignant hyperthermia.
- Family h/o muscular dystrophies.
- Pre-existing hyperkalemia (E.g. : CKD) :
 $\uparrow K^+ \rightarrow$ v.fib $\rightarrow$ Cardiac arrest.
- Chronic liver failure, sepsis.
- Head injury $\uparrow$ ICP/IOP, hiatus hernia.
- Burns, hemiplegia, paraplegia (upto 6 months) :
D/t damage to receptors $\rightarrow \uparrow K^+$ levels $\rightarrow$ Hyperkalemic cardiac arrest.

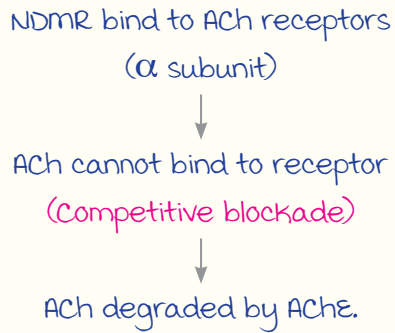
Non-depolarizing Muscle Relaxants

00:26:31

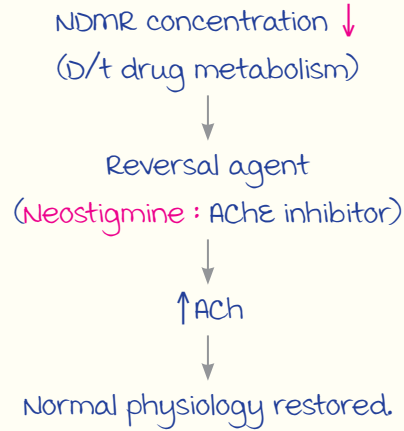
----- Active space -----

MOA :

Start of surgery :



End of surgery :



Steroidal Group :

	Pancuronium	Vecuronium	Rocuronium
Duration & onset	Long acting	Intermediate acting	- Intermediate acting (30 - 45 mins) - Onset : 3 - 4 min 30 s
Features	Sympathetic stimulant	Cardio-stable	Dose : 0.6 - 1.2 mg/kg
Indications	Shock	- Cardiac pt. : AOC - Neuro Sx - GI Sx	- Rapid sequence intubation - Day care surgery : AOC
c/i	- Cardiac patients/ history of MI - CKD (D/t renal metabolism) - Day care surgeries	-	-
Reversal	-	Sugammadex	

----- Active space ----- Reversal agents  :

	Neostigmine	Sugammadex
MOA	AChE $\ominus$	Cyclodextrin molecules : Covalent binding to steroidal NDMR ↓ Renal excretion
Time of administration	EtCO _a : Curare clefts	Anytime
Dose	0.05 - 0.07 mg/kg	2 - 16 mg/kg ↓ Quick reversal
s/e	Bradycardia ↓ Prevent with atropine/ glycopyrrolate	- Anaphylaxis - Contraceptive failure : - Bind to OCPs → Excretion - Alternate contraceptive x 7 - 10 days

Benzylisoquinoline Group (Older Agents)  :

	Atracurium	Cisatracurium (Isomer)	mivacurium
Duration of action	Intermediate acting		Short acting (10 mins)
metabolism	Hoffman's degradation : pH & temperature dependent clearance		Plasma cholinesterases
Advantage	AOC in liver & kidney failure		2 nd preferred in daycare surgery
	-	- No histamine release ↓ Laudanosine - more potent (Preferred)	
Disadvantage	- Histamine release (Anaphylaxis) - Laudanosine production ↓ Crosses BBB ↓ Seizures	-	-

Signs of Adequate Reversal

00:49:44

- Train of four ratio > 0.9 → Safe for extubation (Gold standard).
- Regular respiration with adequate tidal volume.
- Spontaneous eye opening (Unreliable).
- Spontaneous limb movement.
- Able to protrude tongue, cough.
- No cyanosis.
- Able to lift head > 5 sec (most reliable bedside test).
- Able to hold tongue depressor between central incisors.

LOCAL ANAESTHETICS

----- Active space -----

Use : Regional anesthesia.

Classification

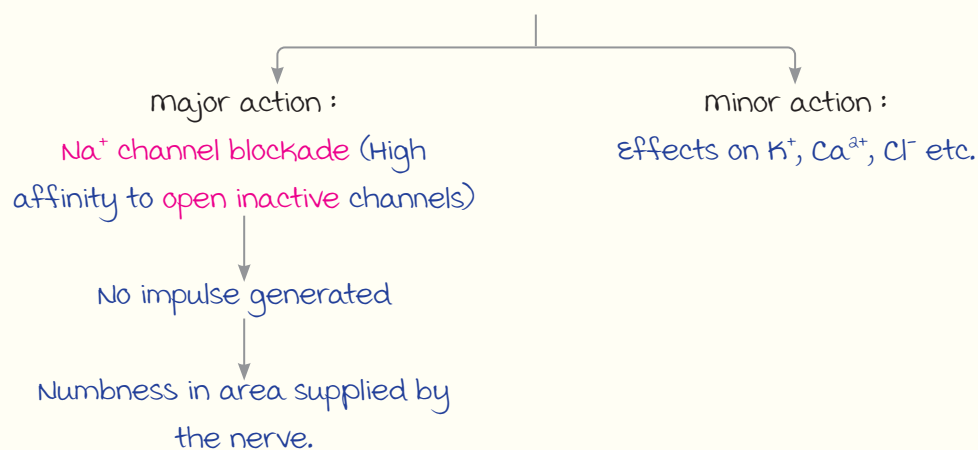
00:00:31

Based on structure :

	Amides	Esters
Stability	Stable	Unstable
Allergic reactions	Less	more
metabolised by	- Liver - Except articaine (Plasma esterases)	- Plasma esterases - Except cocaine (Liver)
Examples	- Lignocaine - Bupivacaine	- Procaine - Cocaine

Mechanism of Action

00:02:44



Components : Based on pH.

	Ionized	Non-ionized
Solubility	Water soluble	Lipid soluble
Entry into cell	⊖	⊕
Activity	Active	Inactive
Na ⁺ channel binding	⊕	⊖

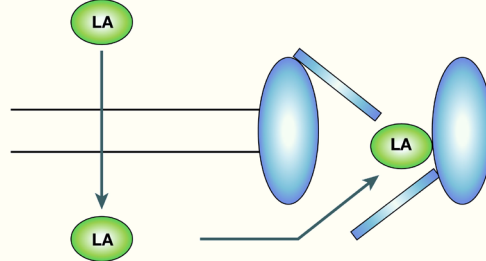
----- Active space -----

Effect of pH on LA :

Henderson Hasselbalch equation : $\text{pH} = \text{pKa} + \log \frac{[\text{Non-ionized}]}{[\text{Ionized}]}$

- $\text{pH} \propto \text{Non-ionized component}$.
- $\text{pH} \propto \frac{1}{\text{Ionized component}}$

1. Unionised form diffused through cell membrane (Basic in nature)



2. Converted to ionised form by lower intracellular pH

3. Reaches target binding site while VGSC is in open state

Action of LA

Clinical significance :

$\downarrow \text{pH}$ in infected areas (Pus, abscess) $\rightarrow \uparrow$ Ionized component $\rightarrow \downarrow$ Entry into the cell $\rightarrow \downarrow$ Action of LA.

Pharmacokinetics

00:11:47

Onset :

Addition of NaHCO_3 (Alkaline medium) $\rightarrow \uparrow$ Non ionized component

$\uparrow$ Entry into cell + conversion

Quicker onset of action.

Absorption :

max systemic absorption :

IV injection > Intercostal nerve block > Epidural > Brachial plexus block.

Duration :

Prolong action by adding :

Adrenaline (vasoconstrictor) :

- 1 : 2,00,000 (5 $\mu\text{g/mL}$)
- $\downarrow$ Bleeding, $\uparrow$ safety margin
- Avoid in end arteries $\rightarrow$ Gangrene.
(Fingertips, Toes, Penis, Nose tip, Pinna)

Opioids :

- Fentanyl.
- morphine.

α_a agonist :

- Clonidine.
- Dexmedetomidine.

Sequence of Blockade

00:18:07

----- Active space -----

Based on myelination :

myelinated nerve fibres > unmyelinated nerve fibres.

Based on size of nerve fibres :

(Erlanger Gasser classification)

- In experimental models : A > B > C.
- In regional/spinal/epidural anesthesia : B > C > A.

Dosage & Toxicity

00:20:44

Dose :

	Dose (mg/kg)
Lignocaine	3 - 4.5
Bupivacaine	2 - 3
Ropivacaine	
Lignocaine + Adrenaline	7

Toxicity :

CC/CNS ratio : $\frac{\text{Cardiovascular collapse}}{\text{CNS toxicity}}$

Drug	CC/CNS ratio	Features 	Rx
Lignocaine	Low	• CNS (1st) : - Circumoral tingling - Visual disturbances - Irrelevant speech - Involuntary limb movement - Seizures • CVS : Cardiac collapse	Stop LA ↓ Inj. midazolam (Short acting BZD)
Bupivacaine & Ropivacaine	High	CVS (1 st) : • VF → Cardiac arrest. • Recovery rate : ↓	20% intralipid : 1.5 mL/kg ↓ F/b 0.25 mL/kg/hr infusion

----- Active space -----

Clinical Applications

00:25:25

Labour & Post-Op Anesthesia (Epidural) :

Drug : Bupivacaine (D/t differential sensitivity).

Dose	Fibre blockade	Effect
Very low (0.0625%)	B	-
Low (0.125%)	C, A δ	Pain relief
High (0.25 - 0.5%)	A α , A β	motor block

EMLA Cream :

- Eutectic mixture of lignocaine & prilocaine : 2.5% (or) 5%.
- Occlusive dressing $\rightarrow$ Surface anesthesia.

Disadvantage :

Large doses of prilocaine in neonates
 $\downarrow$
 methemoglobinemia (Falsely $\downarrow$ SpO_a values).

Uses :

- Painless cannula insertion.
- Dermatological procedures.

Bier's Block :

AKA Intravenous regional anesthesia (IVRA).

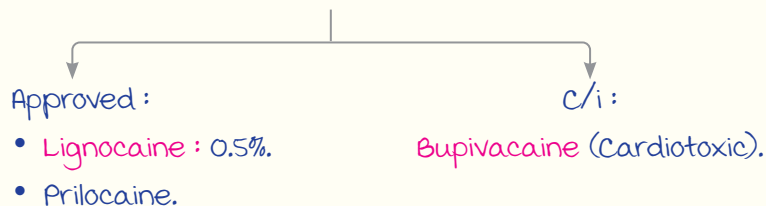
method :

Apply tourniquet $\rightarrow$ Inject LA into vein $\rightarrow$ Regional anesthesia.

Use : Day care procedure.

Disadvantage : Only minor procedures done (D/t tourniquet pain).

Drugs :

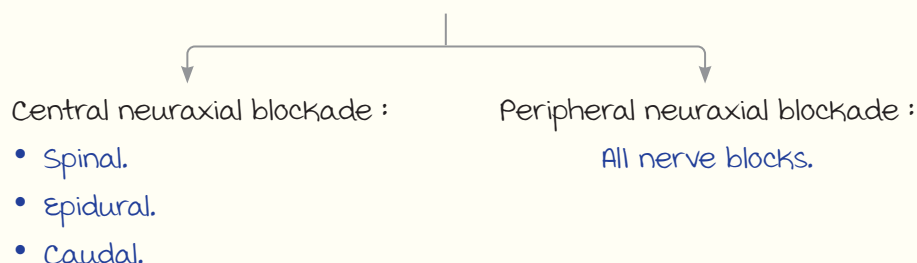
c/i  :

- Sick cell anemia.
 - PAD.
 - AV fistula.
- } Tourniquet application is c/i.

REGIONAL ANAESTHESIA, ANAESTHESIA MACHINES AND CIRCUITS, AND DAY CARE SURGERY

----- Active space -----

Classification :



Spinal Anaesthesia/Sub-Arachnoid Block

00:01:27

Indication :

Sx below level of umbilicus.

Absolute C/I :

- Raised ICP.
- Bleeding disorders.
- Local site infection.
- Severe hypovolemia.
- Severe mitral & aortic stenosis.
- Allergy to the drug.
- Patient refusal.

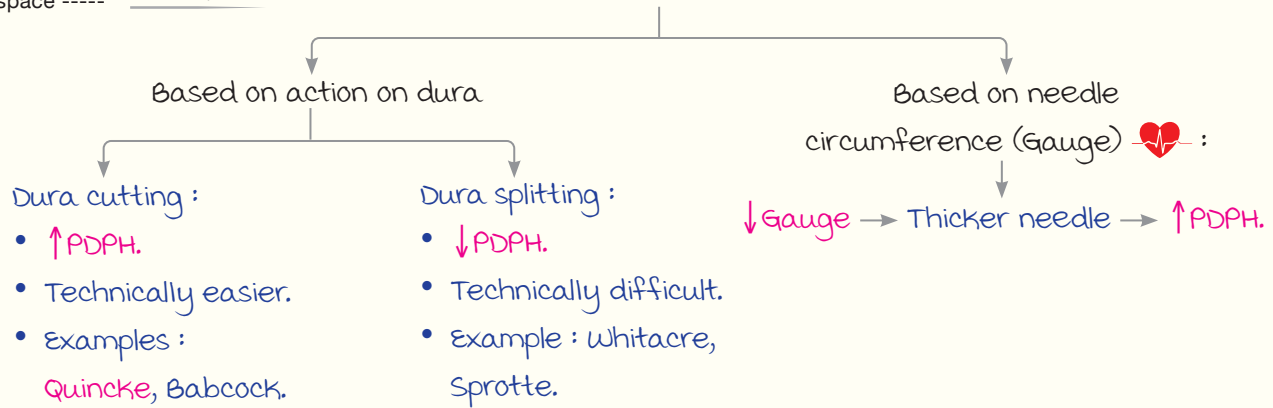
Site :

- Adult : L₃ - L₄.
- Children : L₄ - L₅.

Layers Encountered :

Skin → Subcutaneous tissue → Supra-spinous ligament → Interspinous ligament
 ↓
 Arachnoid ← Duramater ← Ligamentum flavum
 (Toughest layer)

----- Active space ----- Needles :



Post Dural Puncture Headache (PDPH) :

Etiology : CSF leakage.

c/f :

- Young females post LSCS (m/c).
- 24 - 48 hrs after.
- mild - moderate, dull, boring headache.
- ↑es on standing, sitting, walking.
- ↓es on rest.

mx :

- Adequate bed rest + plenty IV fluids.
- Analgesia : Caffeine + paracetamol.
- Severe cases : Epidural blood patch.

Factors affecting Level of Anaesthesia (LOA) :

Patient factors :

- Height $\propto$ 1/LOA.
- CSF $\propto$ 1/LOA.
- Not affected by weight.
- Pregnancy : Higher LOA d/t
 - ↑Intraabdominal pressure → ↑Subarachnoid & epidural venous pressure.
 - Progesterone (Nerves become sensitive).

Drug factors :

- Volume & level of injection $\propto$ Level of anaesthesia.
- Baricity of drug = Density of drug compared to CSF.
 - Hypobaric (Drug floats).
 - Hyperbaric (Drug settles down).

Procedure factors :

----- Active space -----

Position : Related to baricity.

Drug	Trendelenburg (Head down)	Reverse trendelenburg (Head up)
Hyperbaric	Higher level of block	Lower level of block
Hypobaric	Lower level of block	Higher level of block

Side Effects :

Spinal anaesthesia → Sympathetic blockade.

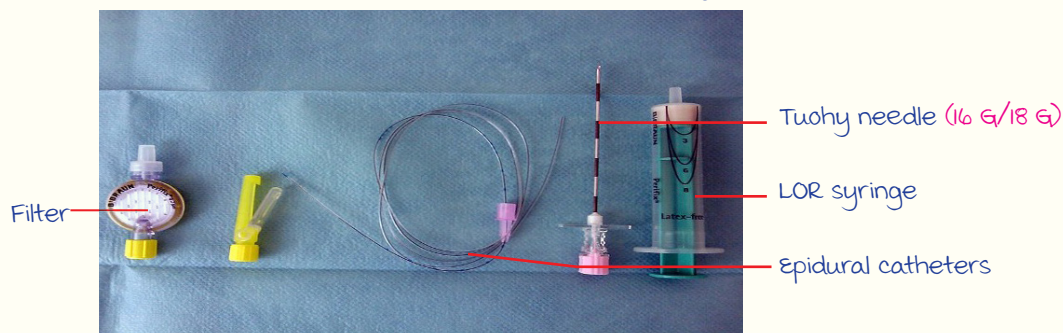
Side effects of spinal anaesthesia	
CVS	↓ HR (Rx : Atropine/glycopyrrolate) ↓ BP : Rx : - Pregnant → Phenylephrine (DOC) - Non-pregnant → Ephedrine (DOC)/mephentermine
Respiratory system	- Low LOA : No effect - High LOA : - Only intercostal muscles paralysed (C/o difficulty in breathing) - Rx : O₂ supplementation
GIT	Sphincters relaxed
GUT	urinary retention (D/t detrusor paralysis) : - m/c in immediate post-op - Rx : Foley's catheter

Epidural Anaesthesia

00:29:02

Technique :

Loss of resistance (LOR) technique (Epidural space : Negative pressure).



Epidural catheter Set

----- Active space -----

Advantages :

- ↑ Duration of anaesthesia.
- Used in post-op analgesia.
- No risk of PDPH (As long as dura is not accidentally punctured).
- Stable hemodynamics.

Disadvantages :

- Technically difficult & not suitable for emergencies.
- Severe PDPH if accidental dural puncture.
- Accidental catheter migration → Subarachnoid space : **Total spinal anaesthesia** (mx : Intubation).
→ Blood vessel : Local anaesthesia toxicity (Seizures/VF).

Note :

High spinal anaesthesia : Diaphragm spared, intercostals paralysed.

Total spinal anaesthesia : **Diaphragm paralysed.**

Caudal Anaesthesia

00:38:18

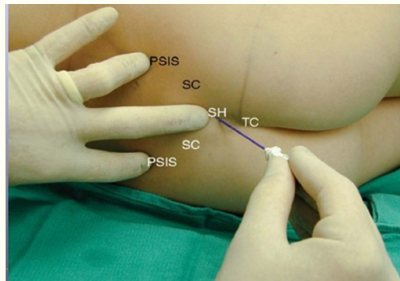
- Done only in children.
- Epidural space extends upto S_a .
- Needle insertion : $S_4 - S_5$ (Unfused sacral vertebrae).

Advantage :

No chance of spinal cord and spinal nerves injury.

Disadvantage :

- Only for post-op analgesia (uncooperative).
- ↑ Chance of infection.



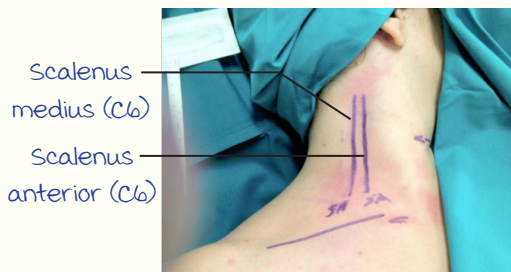
Peripheral Nerve Block

00:41:20

----- Active space -----

Brachial Plexus Block :

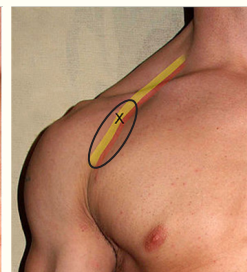
Approach 	Blockage	Indications	Complications/disadvantages
Interscalene	Roots	- upper humerus Sx - Shoulder Sx	100% phrenic block - Pneumothorax - Horner's syndrome
Supraclavicular	Distal trunks (Close to divisions)	- Forearm Sx - Lower humerus Sx	40 - 60% phrenic block - Pneumothorax
Infraclavicular	Cords (Also block musculocutaneous N)	Forearm Sx	Require peripheral nerve stimulator/USG
Axillary	Individual nerves	Forearm Sx	Spare musculocutaneous N



Interscalene NB



Supraclavicular NB



Infraclavicular NB



Axillary NB

Transverse Abdominal Plane (TAP) Block :

Site :

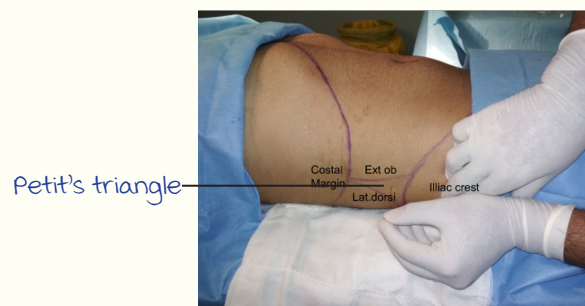
B/w internal oblique and transverse abdominis muscle.

Indication :

Below umbilical Sx.

Disadvantage :

Large dose of anaesthetic required.

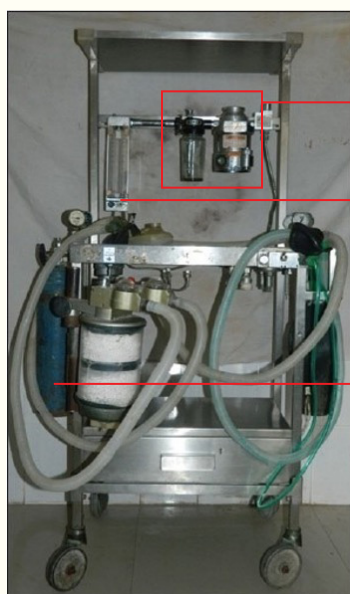


Surface marking for TAP block

----- Active space -----

Anaesthesia Machines

00:51:42

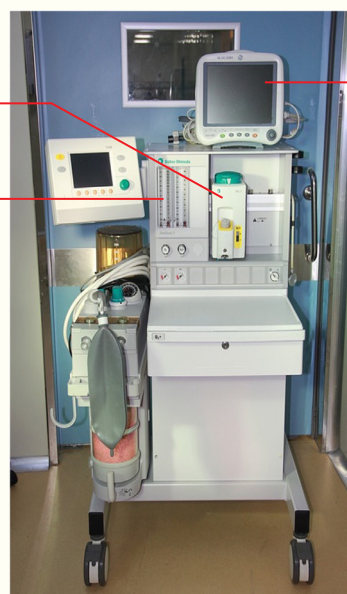


Boyle's machine

Vaporizers

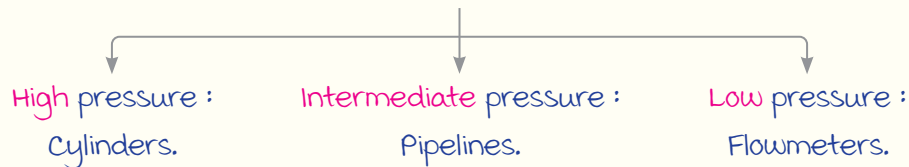
Flow meters

Cylinder

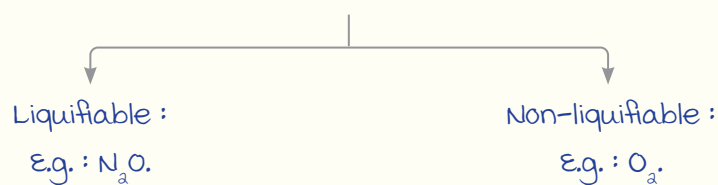


Anaesthesia workstation

monitors

Zones :**Cylinders** ❤️

00:53:22

Classification :

Identification :

----- Active space -----

Based on colour coding system.

Gas	Colour	
O ₂	• Body : Black • Shoulders : white	50%
CO ₂	Grey	Entonox :
N ₂ O	Blue	
Cyclopropane	Orange	50%
Helium	Brown	
Nitrogen	Black	
Air	• Body : white • Shoulders : Black	

Entonox :

- Body : Blue.
- Shoulder : white.
- Labour analgesia (Earlier).

make :

	medical gas cylinders	Special gas cylinders
material	molybdenum steel alloy	Aluminium
usage in MRI rooms	No	Yes

measurement of Pressure :

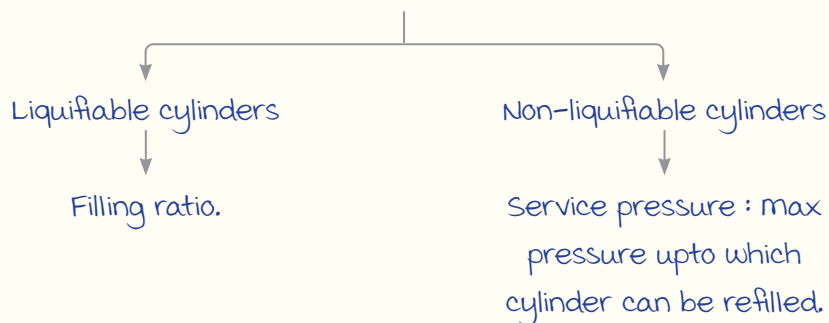
Bourdon's pressure gauge :

Not required for liquifiable gas cylinders.

Gas	Pressure (Pounds/sqr. inch)
O ₂	2000 psi
N ₂ O	760 psi
Cyclopropane	60 psi

Safety Features :

To avoid explosions :



----- Active space -----

To prevent wrong connections - ❤️ :

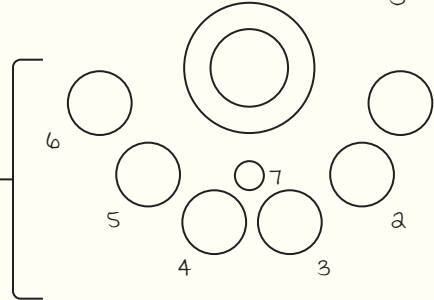
Pin index safety system :



Holes

Left

Right



Cylinder	Pin index
O ₂	2, 5
N ₂ O	3, 5
Air	1, 5
CO ₂ (< 7.5%)	2, 6
CO ₂ (> 7.5%)	1, 6
Entonox	7

To prevent leakages :

Bodock's pressure seal.

- Similar to gasket, ring.
- One pressure seal per cylinder.

Pipelines

01:02:56

Origin : manifold (Storage of cylinders).

material : Copper.

Pressure : 45 - 60 psi.

Diameter Index Safety System :

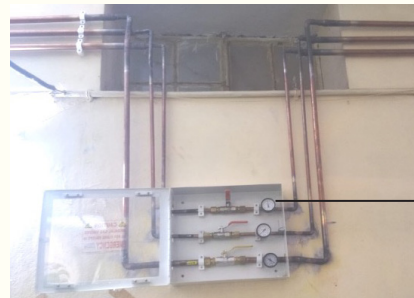
Prevents wrong connection of pipelines.

Oxygen Fail Safe valve :

↓ Pressure inside O₂ cylinder → Automatic ↓ pressure inside N₂O cylinder.

Oxygen Flush valve :

Delivers 35 - 75 L/minute of oxygen.

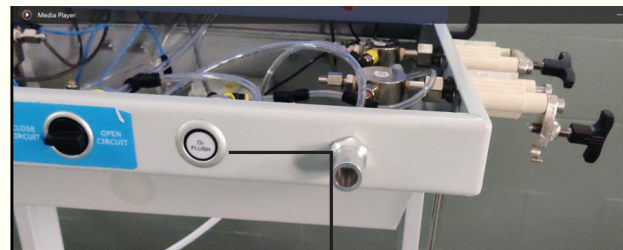


Pipelines

Pipeline pressure indicators



Diameter index safety system



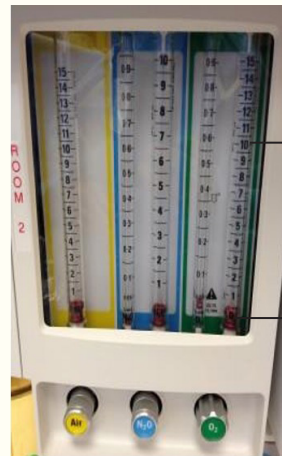
Oxygen flush valve

Flowmeters

01:06:24

----- Active space -----

- Pressure : 10 - 15 psi.
- Constant pressure, variable orifice devices.
- Colour coded.



Flowmeters

O₂ flowmeter :

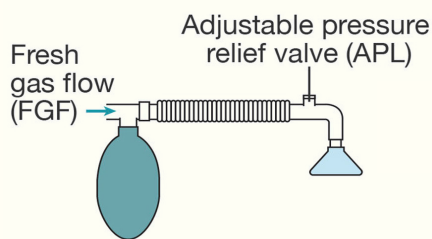
- Always downstream
- Last flowmeter tube

Indicator (Bobbin)

Breathing Systems : Semiclosed/Mapleson System

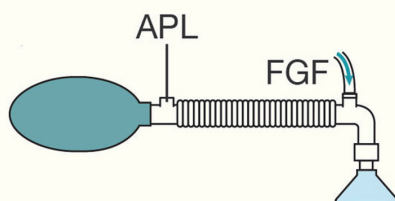
01:10:25

Connection between anaesthesia machine & patient.

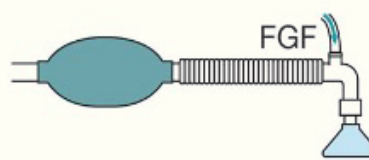


mapleson A

	Features
mapleson A/ magill's circuit	• Spontaneous breathing • modification : Lack circuit
mapleson D	• Circuit for controlled ventilation • modification : Bain's circuit - m/c used - Coaxial circuit
mapleson E	AKA Ayre's T piece
mapleson F	• AKA Jackson Rees circuit • modification of mapleson E • used in children



mapleson D



mapleson F



Bain's circuit

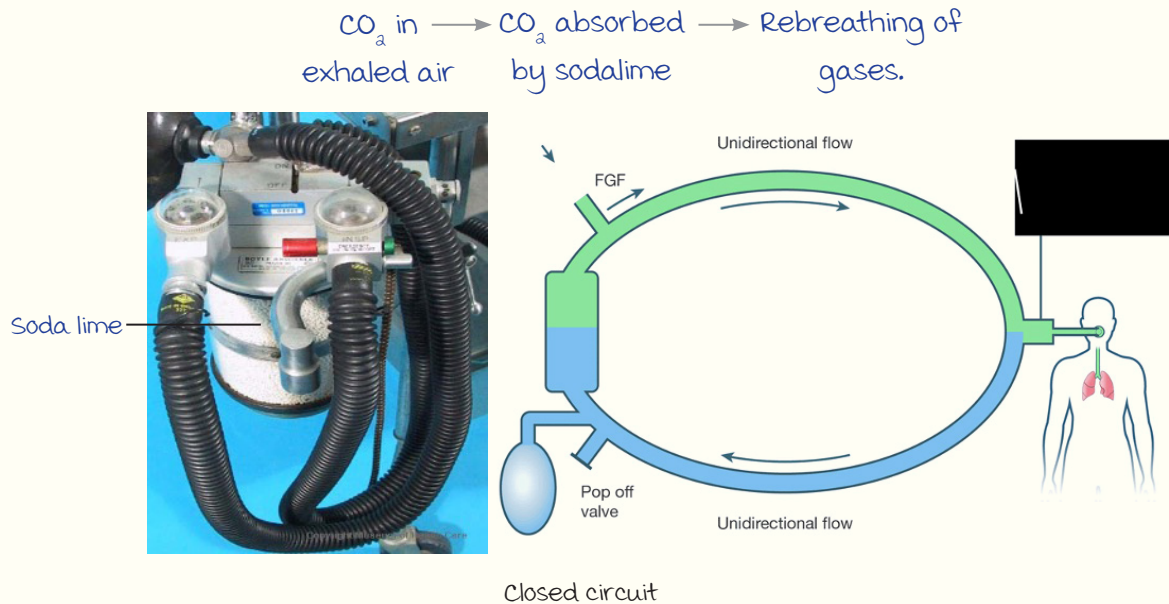
----- Active space -----

Disadvantages :

- Requires large gas flow :
 - used for **transporting** patients.
- Does not conserve humidity.

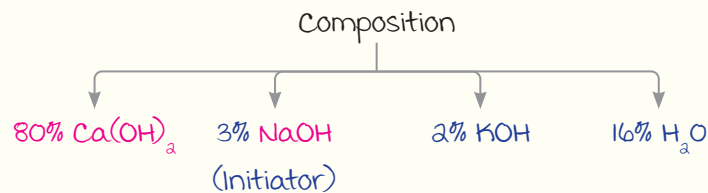
Breathing Systems : Closed Circuits

01:14:34



Advantages :

- Less fresh gas flow.
- Humidity conserved.

Disadvantage : **Bulky** circuit ($\uparrow$ Chances of disconnection).Sodalime :

Reaction : Exothermic.

- Step 1 : $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$.
- Step 2 : $\text{H}_2\text{CO}_3 + \text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$.
- Step 3 : $\text{Na}_2\text{CO}_3 + \text{Ca(OH)}_2 \rightarrow \text{CaCO}_3 + 2\text{NaOH}$.

Disadvantages :

- Soda lime + sevoflurane $\xrightarrow{\text{Prolonged use } (> 48 \text{ hrs})}$ **Compound A** (Nephrotoxic).
- Soda lime + inhalational agents $\rightarrow \text{CO}$.
- Baralyme + desflurane $\rightarrow \text{maximum CO}$.

Day Care Surgery

01:22:07

----- Active space -----

Patient Selection :

- ASA I, II : Ideal.
 - ASA III : Considered for minor Sx.
 - Premature babies.
 - > 85 yrs.
- } Avoided.

Procedure :

- mild - moderate Sx.
- upto 90 min.
- minimally invasive Sx.
- Not anticipating complications.

Anaesthetic Conditions :

- PAC.
 - Fasting.
 - monitoring.
- } Same as before.

Drugs :

Short acting : No residual effects.

Agent	Drug
IV induction agent	Propofol
Inhalational agent	Sevoflurane > Desflurane
muscle relaxant	Rocuronium + Sugammadex
Opioid	Remifentanyl ; Fentanyl
Regional anesthesia	Can be done
Local anesthetic	Chloroprocaine (Shortest acting LA)

Airway : 2nd gen LMA > 1st gen LMA.

Discharge criteria :

modified Aldrete scoring : > 9 → Fit for discharge.

Complication  :

- Nausea & vomiting.
 - Drowsiness.
 - Pain.
- m/c

Re-admission : Bleeding (m/c cause).

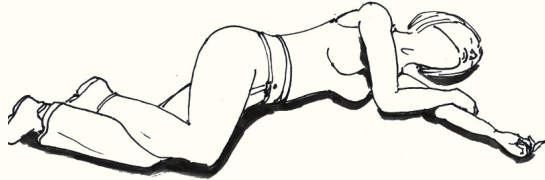
----- Active space -----

AIRWAY MANAGEMENT

Techniques :

Left lateral position  :

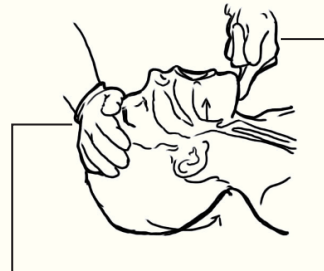
- Unconscious.
- Pulse : $\oplus$.
- Breathing : Normal.



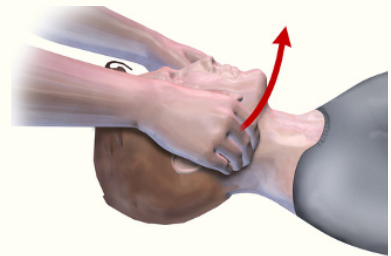
Left lateral position

Triple manoeuvre  :

- Head tilt.
- Chin lift.
- Jaw thrust.



Head tilt



Chin lift

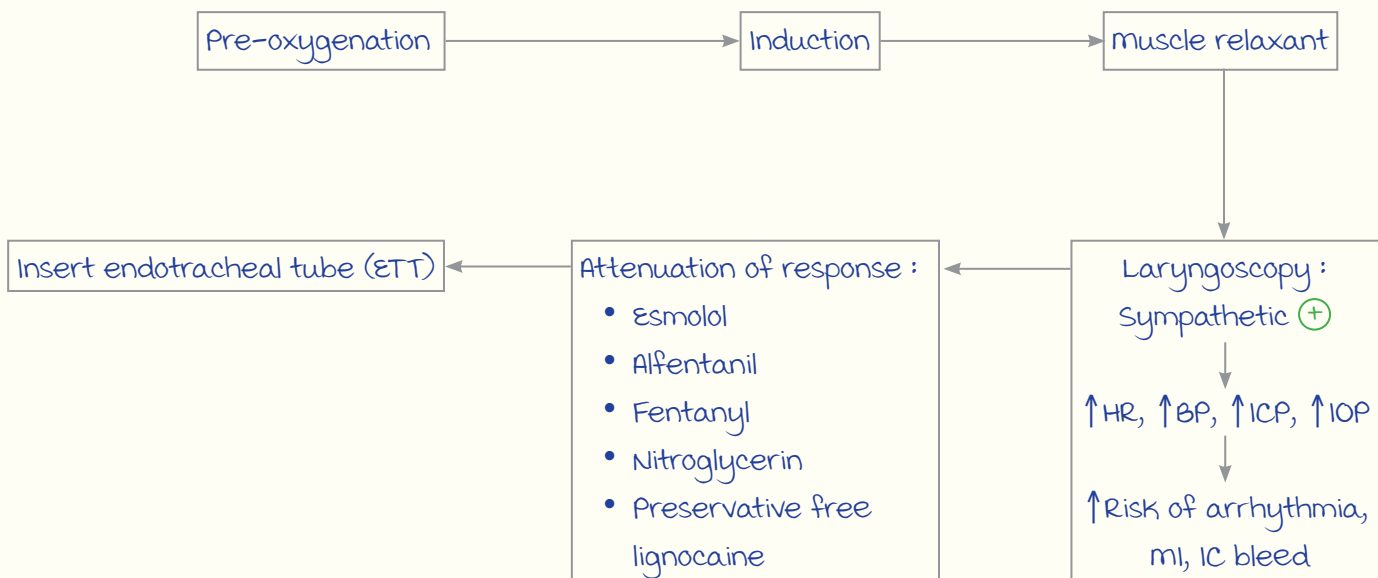
Intubation :
Gold standard.

Jaw thrust

Intubation

00:03:28

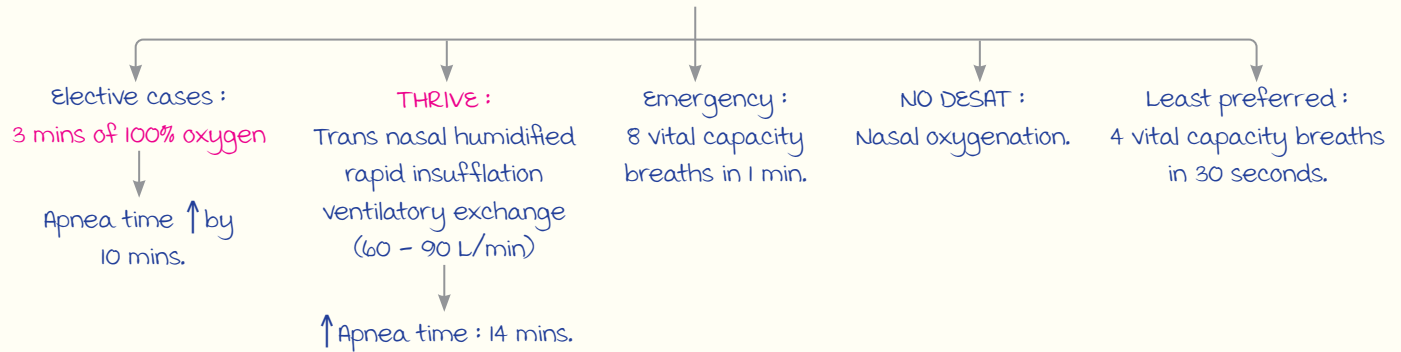
Steps :



Pre-oxygenation :

----- Active space -----

- Provide 100% O_a : ↑ Apnea time.
- Normal apnea time : 3 mins (↓ In obese, smokers, LRTI).

methods :Technique :

- Triple manoeuvre.
- C & E technique.



Anatomical face mask



C & E technique

Airway Instruments :Guedel's airway :

- Prevent tongue fallback.
- ↓ Tolerance (Triggers gag reflex).
- Require deep sedation.

Nasopharyngeal airway -❤️- :

- Prevent pharyngeal collapse.
- Better tolerance.



----- Active space -----

Laryngoscopy

00:13:26

Blades :

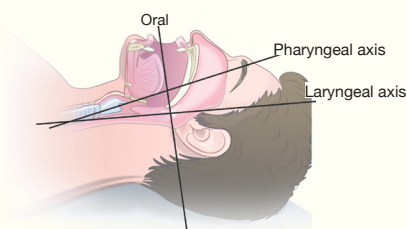
	macintosh blade Curved blade	millers blade Straight blade
Type		
Age group	Adults	Children
Procedure ❤️	Insert from right corner of mouth with left hand ↓ Push the tongue ↓ Advance into oral cavity ↓ Reach base of tongue (Epiglottis visualized) ↓ Lift hand with arm (Deltoid & triceps) w/o including epiglottis	Insert from centre of oral cavity ↓ Depress the tongue ↓ Advance into oral cavity ↓ Lift hand with arm (Deltoid, bicep) including epiglottis

Position ❤️ :

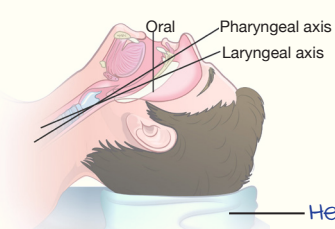
Align oral, pharyngeal & laryngeal airways.

Sniffing position :

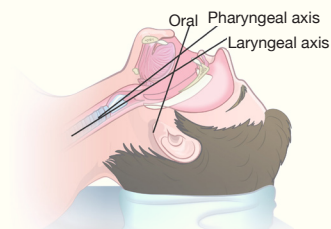
- Lower C spine : Flexion.
- Atlanto occipital joint : Extension.



3 axes in neutral position



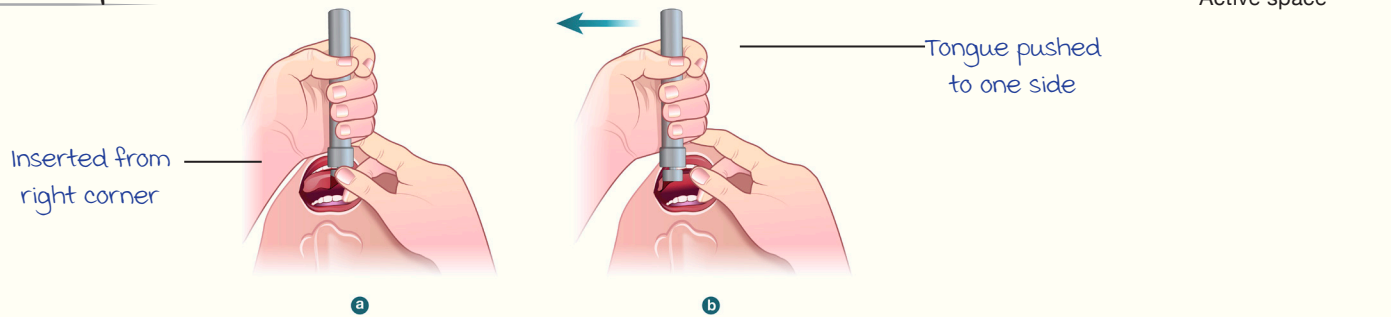
Sniffing position



Triple manoeuvre

Note :

Rose position : Tonsillectomy.

Technique :Cormack Lehane Scoring :

Post-laryngoscopic assessment of glottic opening.

Grade I	Grade II	Grade III	Grade IV
Epiglottis Vocal cord Entire glottic opening	Posterior portion of laryngeal aperture	Only epiglottis	No epiglottis

Endotracheal Tubes

00:21:25

Cuffed Endotracheal Tubes :

Types :

High pressure,
low volume.

<

Low pressure,
high volume.

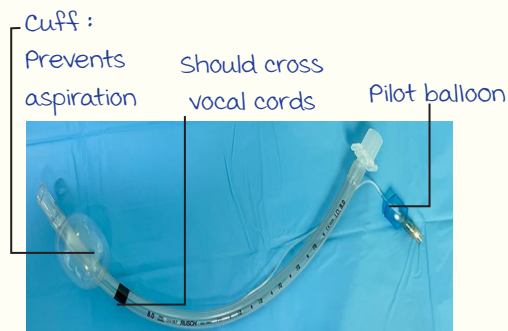
Size :

- male : 8.0, 8.5, 9.0 mm
- Female : 7.0, 7.5, 8.0 mm
- Children :
 - < 6 yrs : $\frac{\text{Age}}{3} + 3.75$
 - > 6 yrs : $\frac{\text{Age}}{4} + 4.5$
 - Length : $\frac{\text{Age}}{2} + 12 \text{ cm.}$

Internal diameter (ID)

- Smallest :

	Preterm	Term	upto 1 yr
ID (mm)	2.5	3.0	3.5
Length (cm)	7	8	9

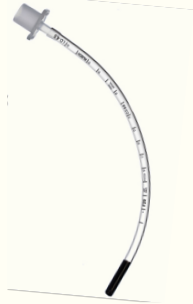


----- Active space -----

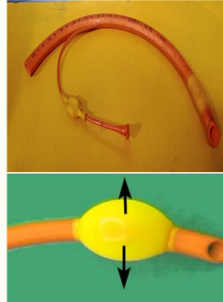
Other ETTs :

1. Uncuffed ETT :

Children : Narrowest position
of larynx is subglottis.



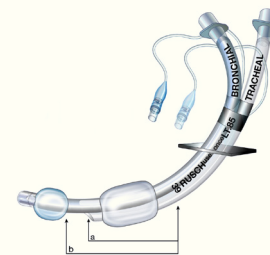
2. High pressure low volume cuffed ETT.

3. microcuffed ETT :
Recent : Children.modifications of ETTs :

1. Flexometallic ETT :

Avoid kinking :

- Head and neck surgery.
- Prone position surgery.

2. Double lumen ETT :
Lung surgery.

3. North-facing R.A.E ETT :

Through nasal route :

- Jaw surgery.
- Oral surgery.



4. South-facing R.A.E ETT :

- Cleft lip surgery.
- Ocular surgery.



Difficult Airway Management

00:29:36

----- Active space -----

Laryngeal mask Airway/Supraglottic Airway :

Classic LMA :

- Advantage :
 - Considered in difficult intubation.
 - Used in emergency, day care Sx.
 - Doesn't require expertise.
 - minimal neck movement.
 - Less invasive.
- Disadvantage :
 - Cannot prevent aspiration : As above esophagus.
 - ↑ Risk of laryngospasm while removal.

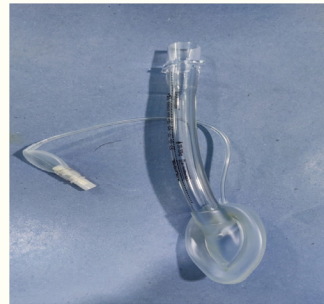
• Avoided in :

- Pregnancy.
- Laparoscopic Sx.
- Prone position.
- Hiatus hernia.
- Full stomach.

Classic LMA :
1st generation.



LMA unique :
1st gen PVC LMA.



Fast-track/intubating LMA :

- 1st generation.
- Can pass ETT.



Proseal LMA :

- Drain tube : 2nd generation.
- ↓ Aspiration.



Ambu LMA 40  :

- 1st generation.
- Auragain :
2nd gen Ambu LMA.



LMA supreme :
2nd gen.



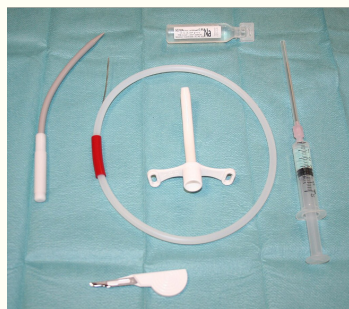
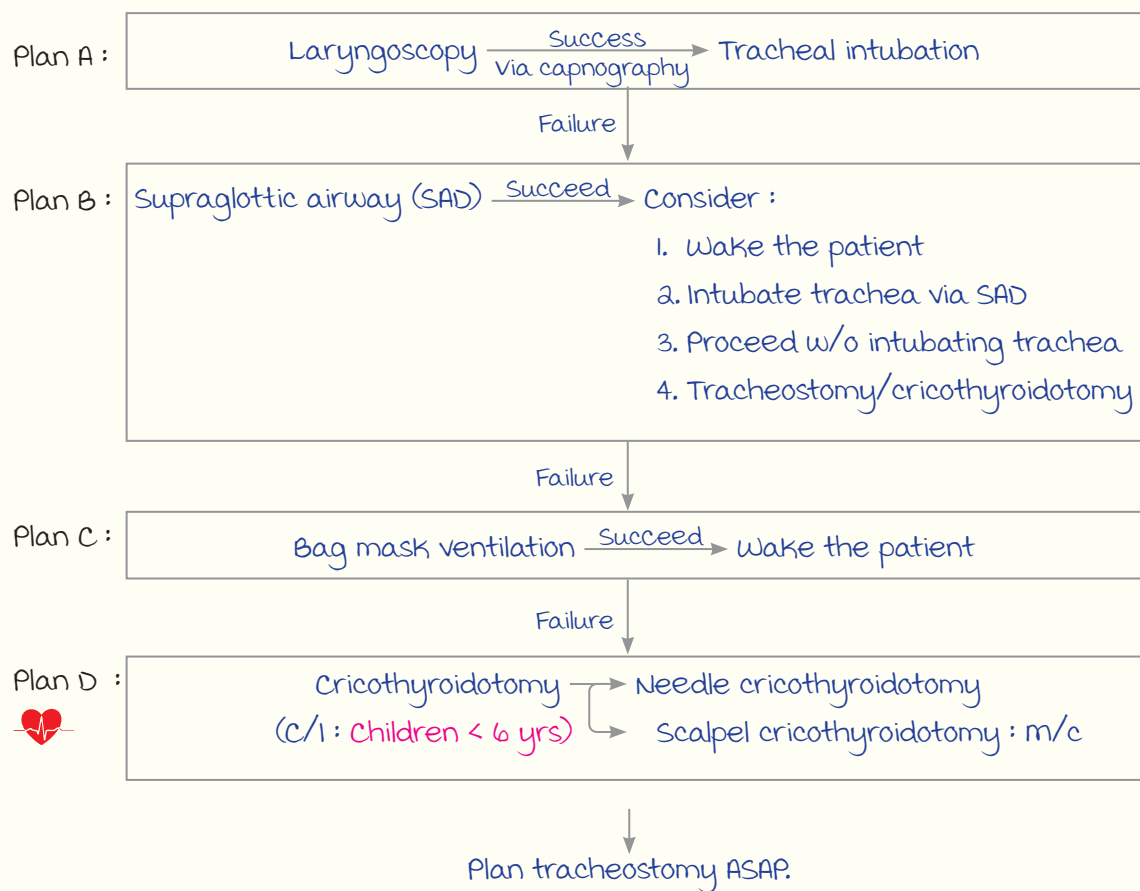
I-GEL :

- 2nd gen.
- No pilot balloon.
- Silicone gel.

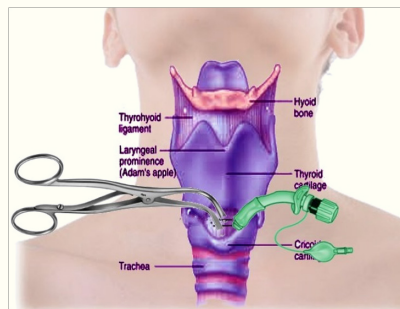


- Better sealing pressure.
- Considered in pregnancy & laparoscopic Sx.

----- Active space -----

DAS Difficult Intubation Guidelines :

Needle cricothyroidotomy



Scalpel cricothyroidotomy

modifications of Intubation :**I. manual inline stabilization of neck (MILS) :**

- Suspected cervical spine injury.
- Prevent neck movement.



2. Rapid sequence intubation (RSI) :

- SELICK's manoeuvre :
 - Cricoid pressure :
Occlude esophagus → ↓ Aspiration.
 - Full stomach patients : RSI.
- Drugs : Thiopentone sodium + succinylcholine.



----- Active space -----

Note :

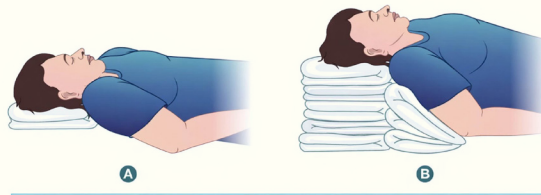
RTA : Rule out pneumothorax → ICD decompress → MILS intubation.

modified RSI :

- Gentle PPV allowed.
- Drugs : Propofol + rocuronium.

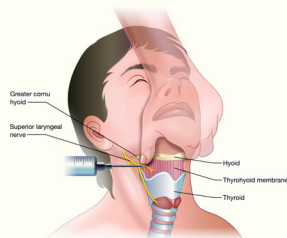
Ramp position :

- Obese patients.
- Imaginary line from ear passes through sternum.

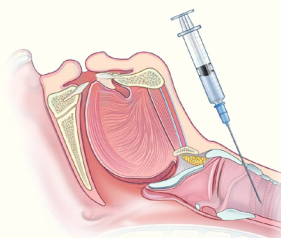


Awake intubation :

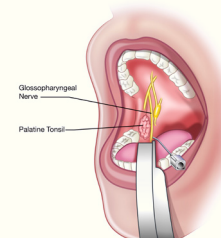
- Done in case of large tumor, Restricted opening of mouth.
- Assisted by nerve blocks.



Superior laryngeal nerve block (External branch)



Recurrent laryngeal nerve block :
Transtracheal injection



Glossopharyngeal nerve block

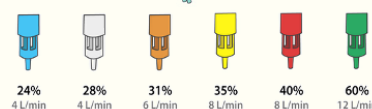
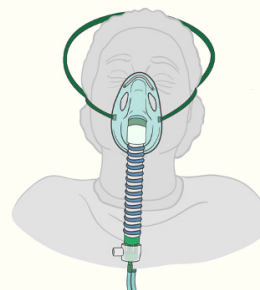
Oxygen Delivery Devices

00:50:00

High Flow Fixed Performance :

Venturi mask :

- Delivers fixed O_2 concentration.
- COPD, Type II respiratory failure.



venturi mask

----- Active space ----- Low Flow Variable Performance :

	Nasal prongs	Simple O ₂ mask	Non-rebreathing mask
			
Flow rate	1 - 6 L/min	6 - 10 L/min	10 - 15 L/min
maximum O ₂ delivered	45%	60%	85%
Features 	Each 1 L gas flowed ↑ FiO ₂ by 4%	-	Has reservoir bag

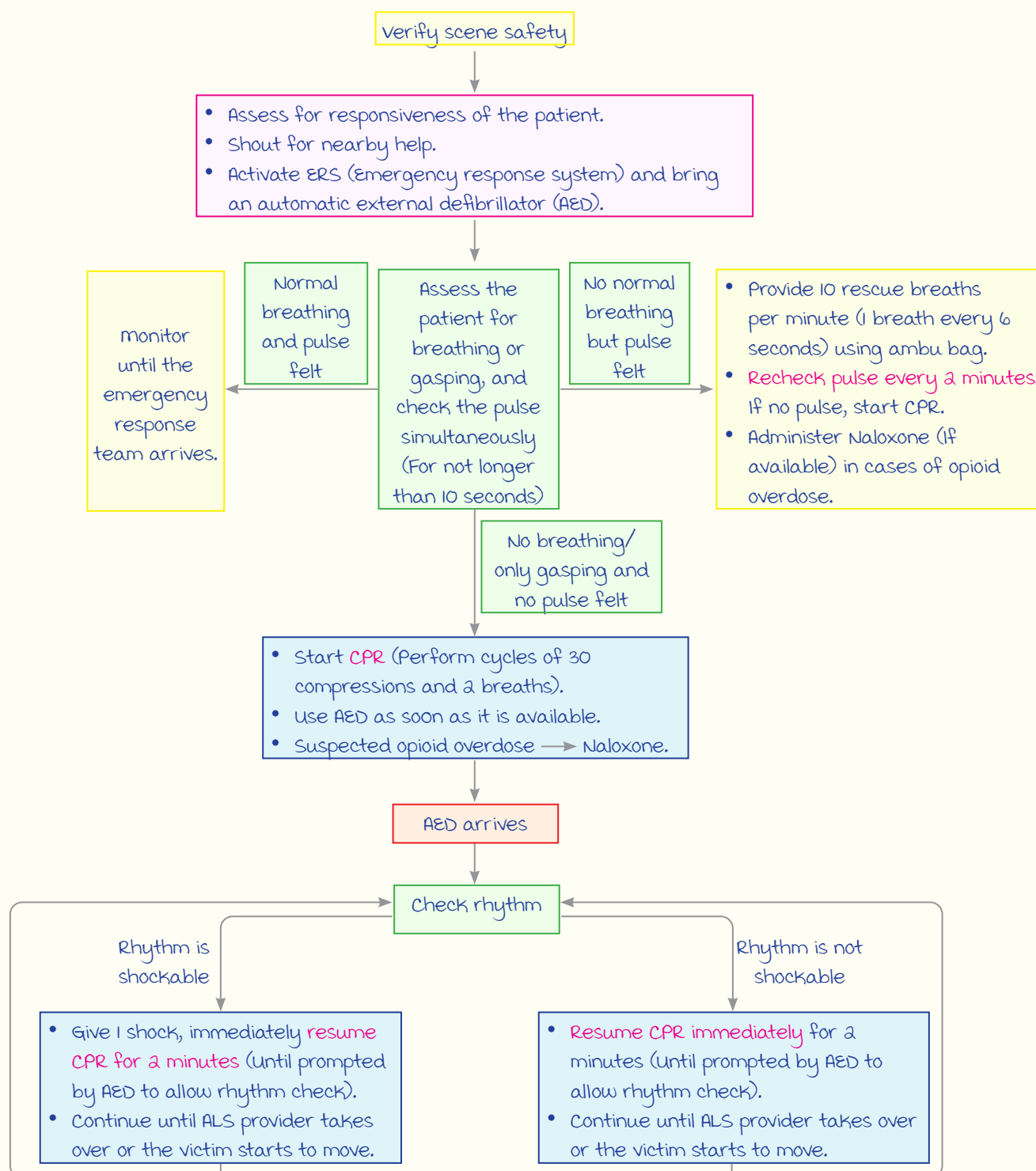
BLS AND ACLS

----- Active space -----

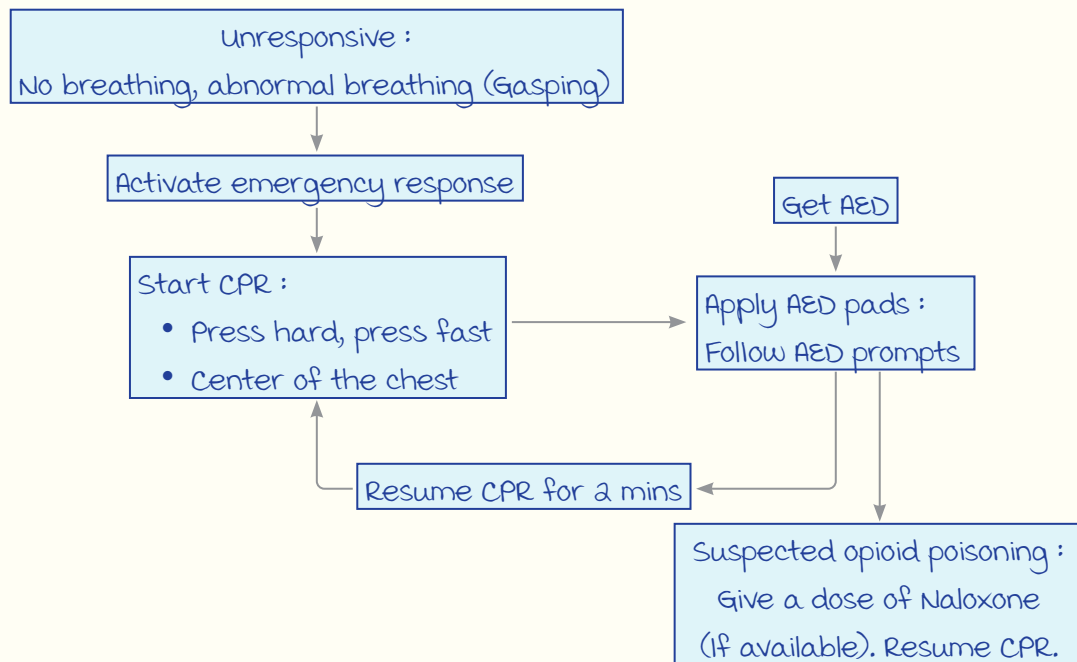
ACLS & BLS guidelines 2025 : American college of cardiology & American heart association.

Basic Life Support (BLS)

00:02:08



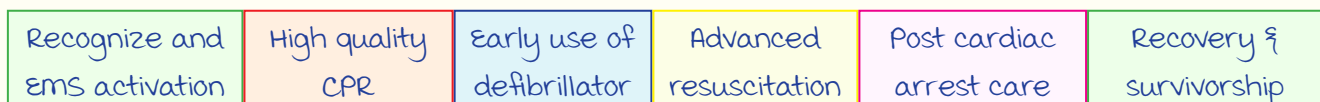
----- Active space -----

Simplified BLS :updates :

- Simplified, user-friendly.
- Do on firm surface at level of rescuer's knee.
- Can adjust brassiere in female while applying AED pads.
- Can do triple maneuver :
 - If jaw thrust ineffective.
 - Even in trauma.
- Same protocol in obese.
- No use of mechanical CPR devices.
- Naloxone can be given by bystanders.

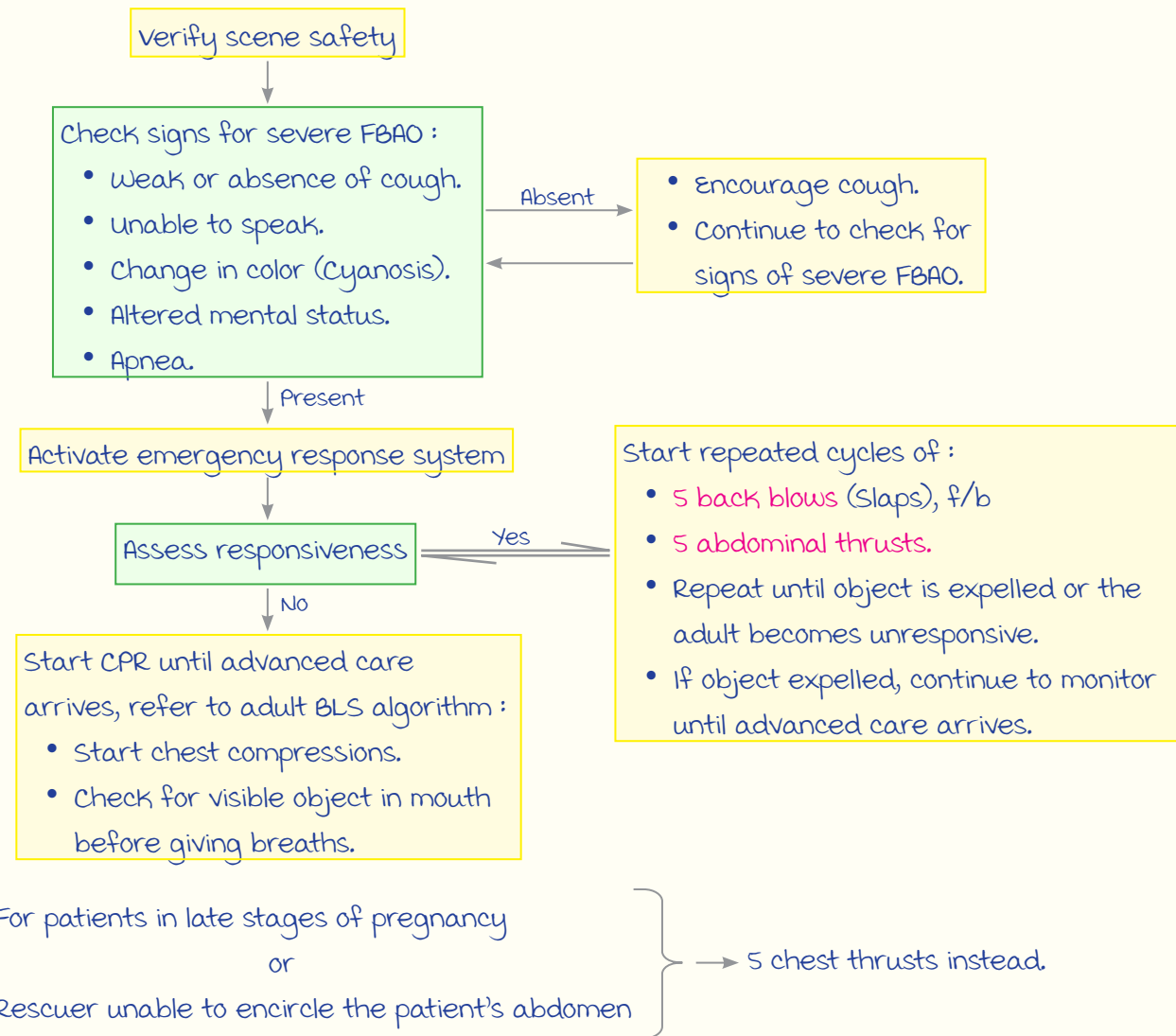
One Chain of Survival :

Same for adult/child, inside/outside hospital cardiac arrest.



Adult Foreign Body Airway Obstruction Algorithm :

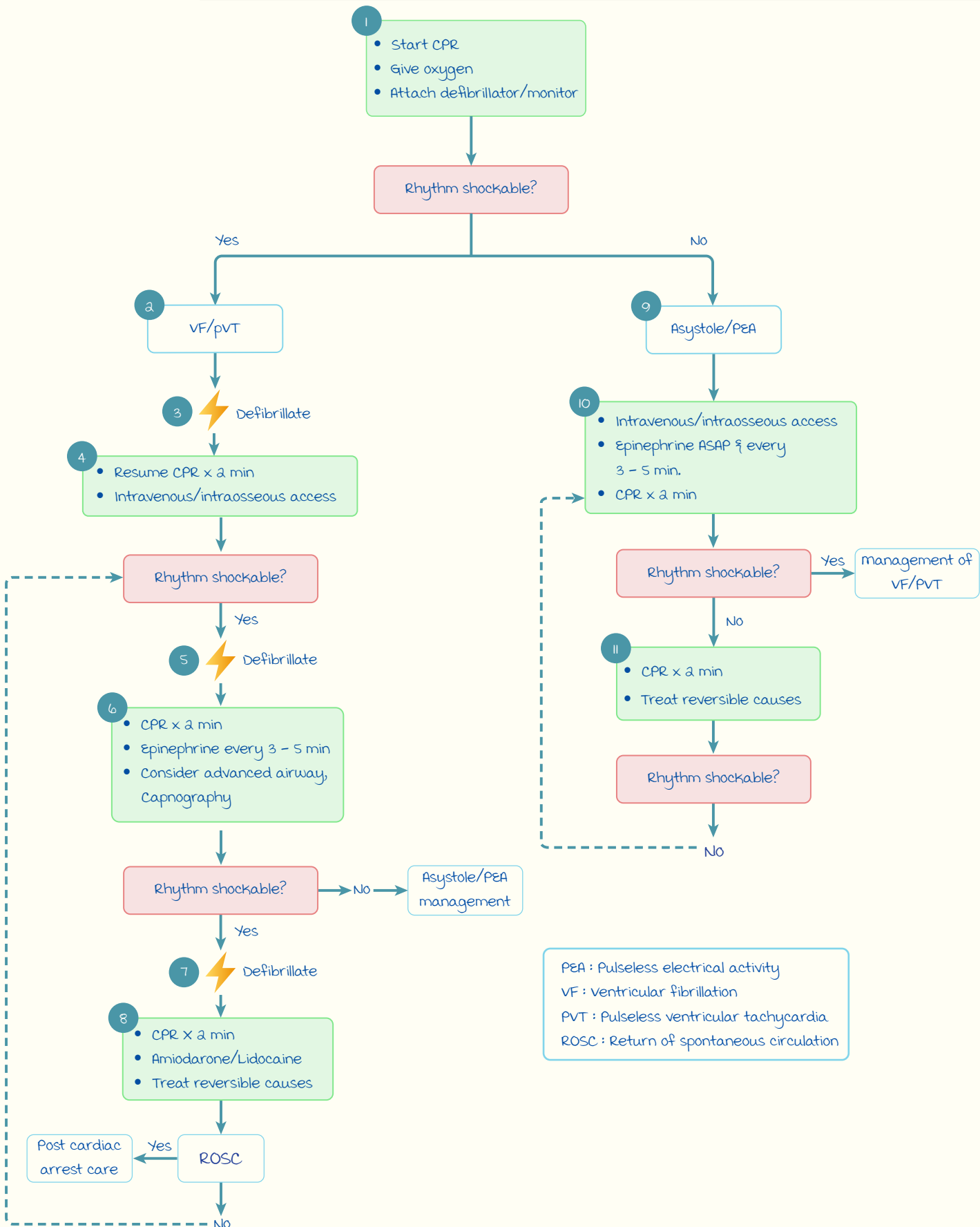
----- Active space -----



----- Active space -----

Advanced Cardiac Life Support (ACLS)

00:17:18



High quality CPR :

----- Active space -----

- Push hard : Atleast **2 inches/5 cm**.
- Push fast : **100 - 120/min** & allow complete chest recoil.
- minimize interruptions in compression.
- Avoid excessive ventilation.
- Change compressor every 2 min or sooner if fatigued.
- If no advanced airway → use **30 : 2** compression ventilation ratio.
- If advanced airway in place → 1 breath **every 6 seconds** (10/min) with continuous compressions.
- CPR quality : Continuous waveform EtCO_a.

Shock energy for defibrillation :

- Biphasic :
 - **120 - 200 J**.
 - If unknown → max. dose.
 - Subsequent doses : Equivalent/Higher.
- monophasic : **360 J**

Drug therapy :

- Epinephrine IV/IO dose : 1 mg every 3 - 5 minutes.
- Amiodarone IV/IO doses :
 - First dose : 300 mg bolus.
 - Second dose : 150 mg. (or)
- Lidocaine IV/IO :
 - First dose : 1 - 1.5 mg/kg.
 - Second dose : 0.5 - 0.75 mg/kg

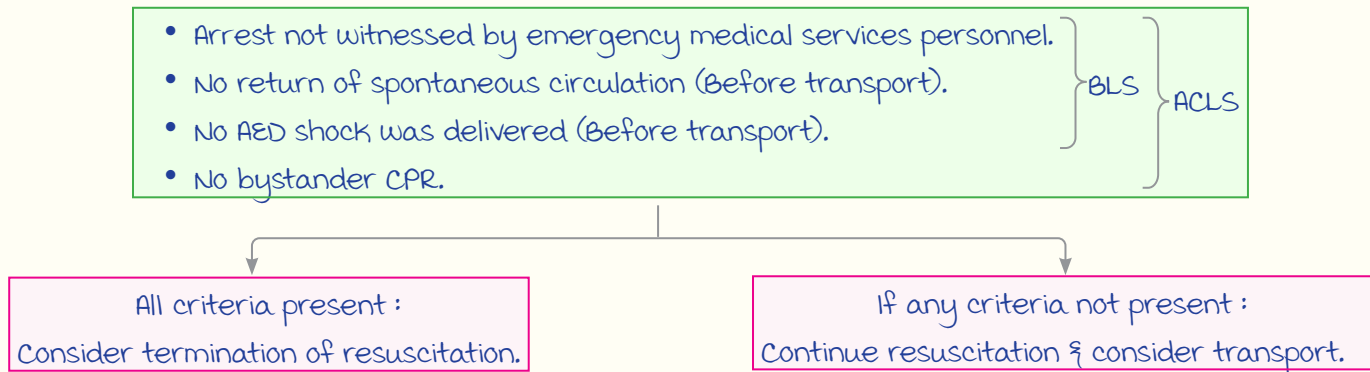
Reversible causes :

- | | | |
|---|---|-------------|
| <ul style="list-style-type: none"> • Hypovolemia. • Hypoxia. • Hydrogen ion. (Acidosis) • Hypo/hyperkalemia. • Hypothermia. | } | 5 Hs |
| <ul style="list-style-type: none"> • Tension pneumothorax. • Cardiac Tamponade. • Toxins. • Thrombosis. (Pulmonary) • Thrombosis. (Coronary) | } | 5 Ts |

Update :

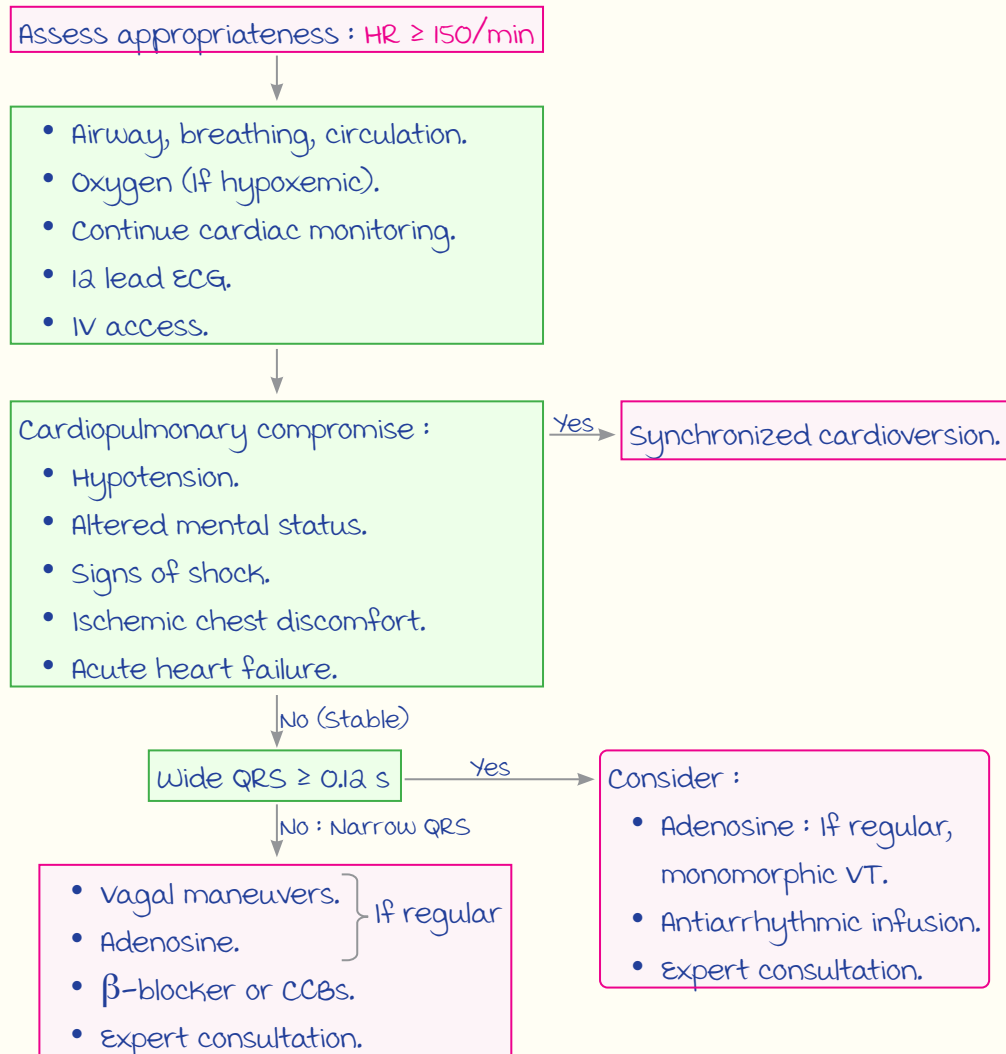
- Simplified guidelines.
- No role of vector change defib.
- No role of head up CPR.
- **IV > IO**.
- No role of β -blockers, sotalol, procainamide in VF/pVT.

----- Active space ----- **ACLS & BLS Termination of Resuscitation :**



Adult Tachyarrhythmia with Pulse Algorithm

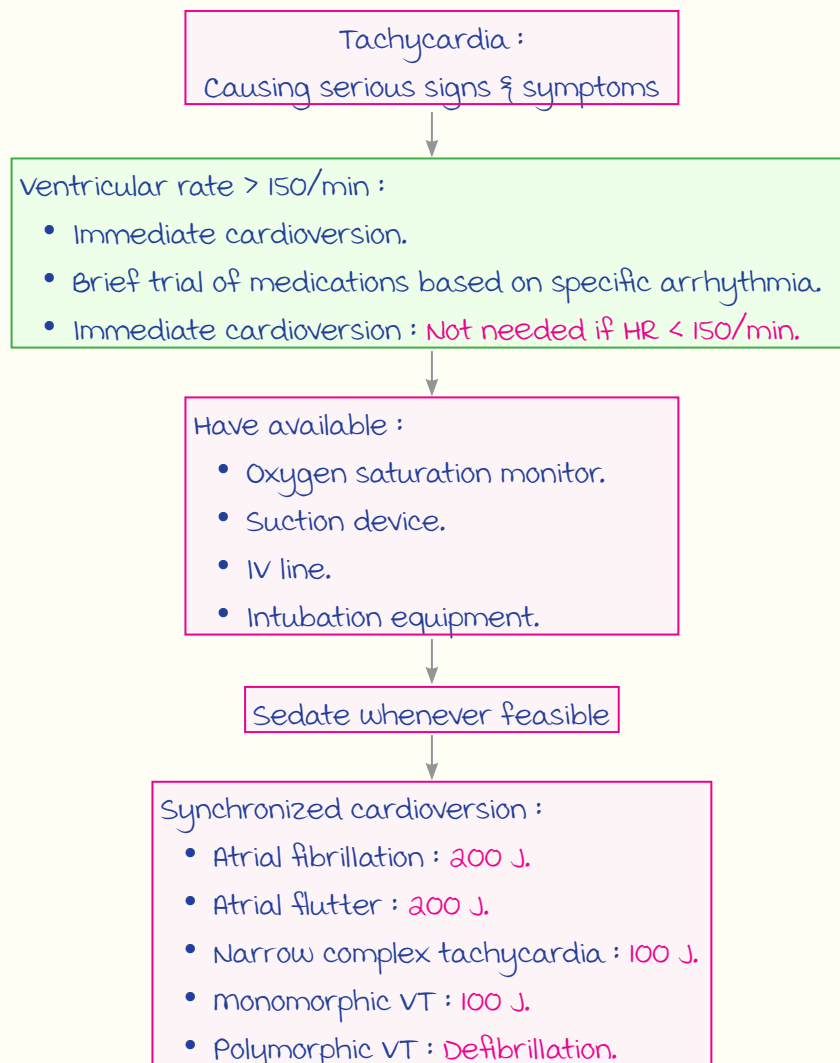
00:24:01



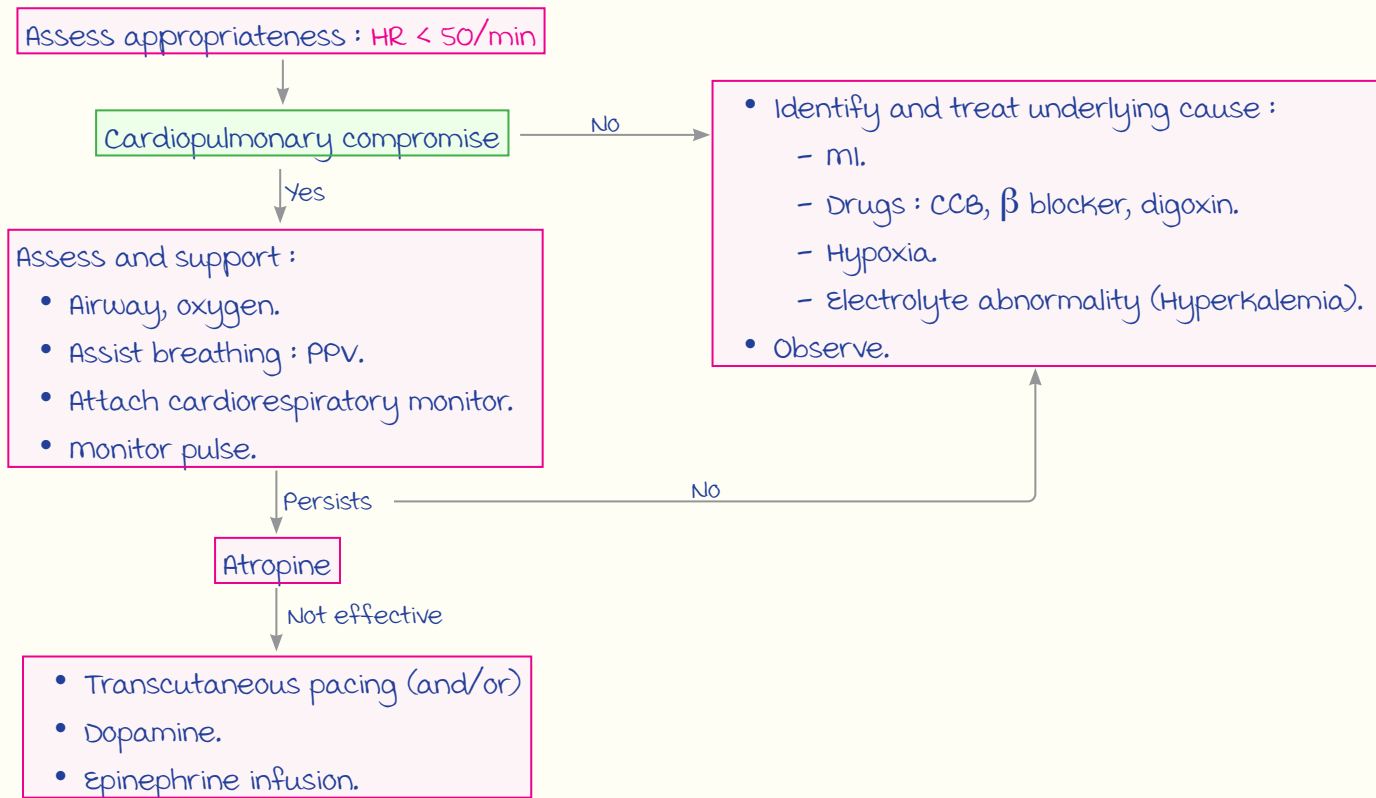
Antiarrhythmic infusion :

----- Active space -----

- Procainamide :
 - 20 - 50 mg/min until arrhythmia is suppressed.
 - Maximum dose : 17 mg/kg.
 - Maintenance infusion : 1 - 4 mg/min.
 - Avoid if prolonged QT or CHF.
- Amiodarone :
 - First dose : 150 mg over 10 mins.
 - Repeat if VT recurs.
 - Maintenance : Infusion of 1 mg/min for first 6 hrs $\xrightarrow{F/b}$ 0.5 mg/min for next 18 hrs.
- Adenosine :
 - First dose : 6 mg rapid IV push then NS flush (Peripheral line).
 - Second dose : 12 mg.

Electrical Cardioversion Algorithm :

----- Active space -----

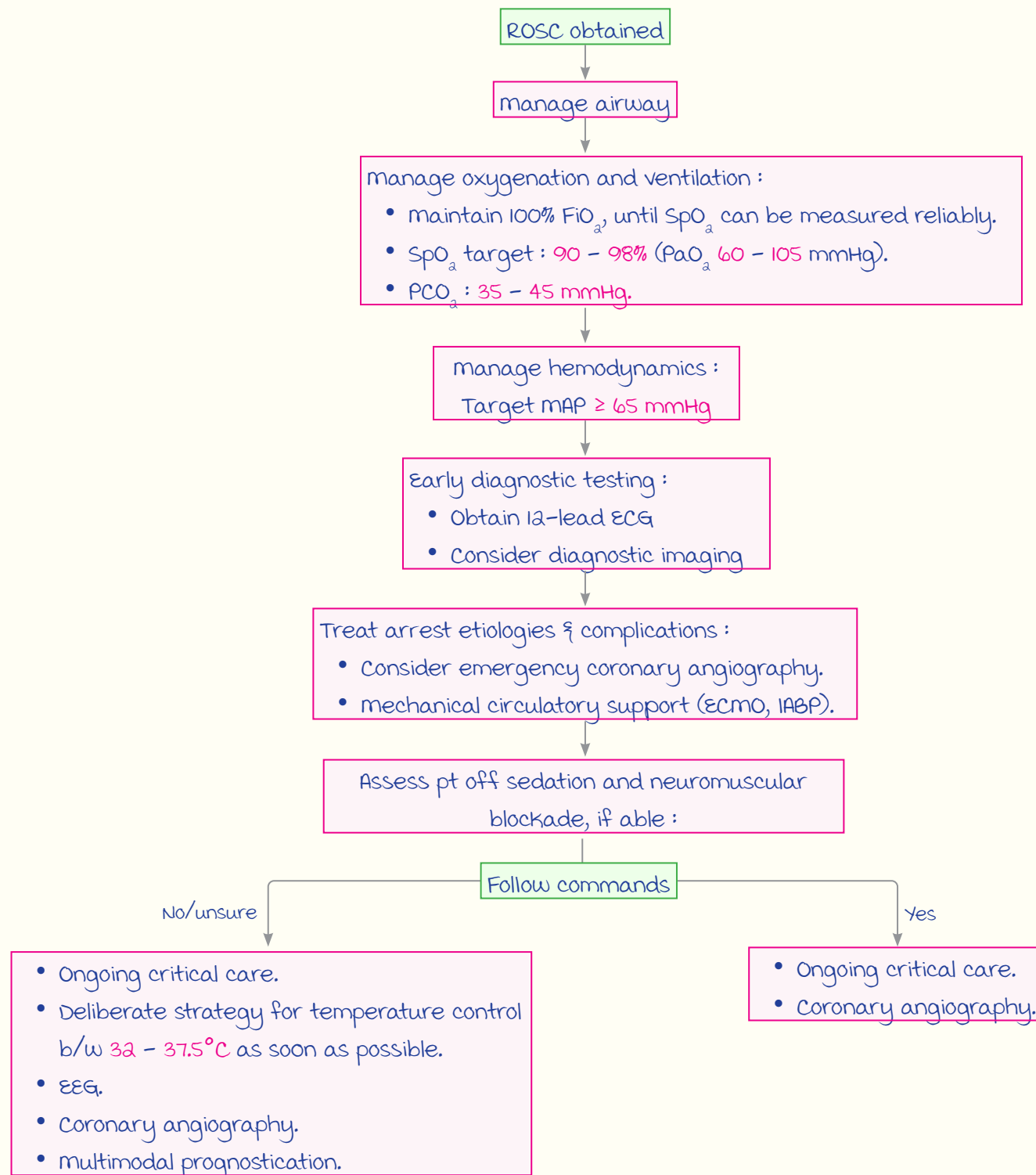
Adult Bradycardia with Pulse Algorithm**Drug dose :**

- Atropine :
 - 1st dose : 1 mg bolus.
 - Repeat every 3 - 5 mins.
 - max dose : 3 mg.
- Dopamine : 5 - 20 mcg/kg/min.
- Epinephrine : 2 - 10 mcg/min infusion.

Adult Post Cardiac Arrest Care Algorithm

00:34:52

----- Active space -----



----- Active space -----

- Neuroprognostication is key.
- **Neurofilament light chain** : New biomarker.
- Avoid hypotension : MAP > 65 mmHg.
- Get CT/POCUS.
- Get CAG (if recovered).
- Reasonable to continue targeted hypothermia upto 72hr.
- Do not treat myoclonus without EEG
- Emotional support from family.

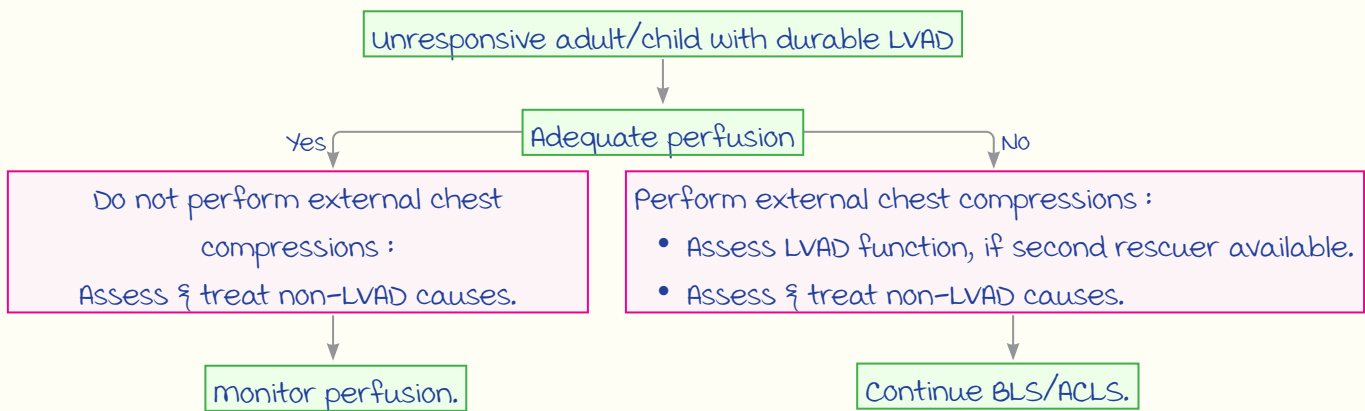
Neuroprognostication:

ROSC			
CT:	Reduced greywhite ratio (GWR)		
	mRI:	Extensive Apparent diffusion coefficient	
		SSEP:	Burst suppression
EEG:	Continuous		
			- Pupillary reflex ⊕ - Good withdrawal to pain
Biomarkers:	↑NSE, NFL		
24 hrs	48 hrs	72 hrs	

- Unfavourable.
- Favourable.
- Somatosensory evoked potential : SSEP.

Adult and Pediatric Durable LVAD Algorithm

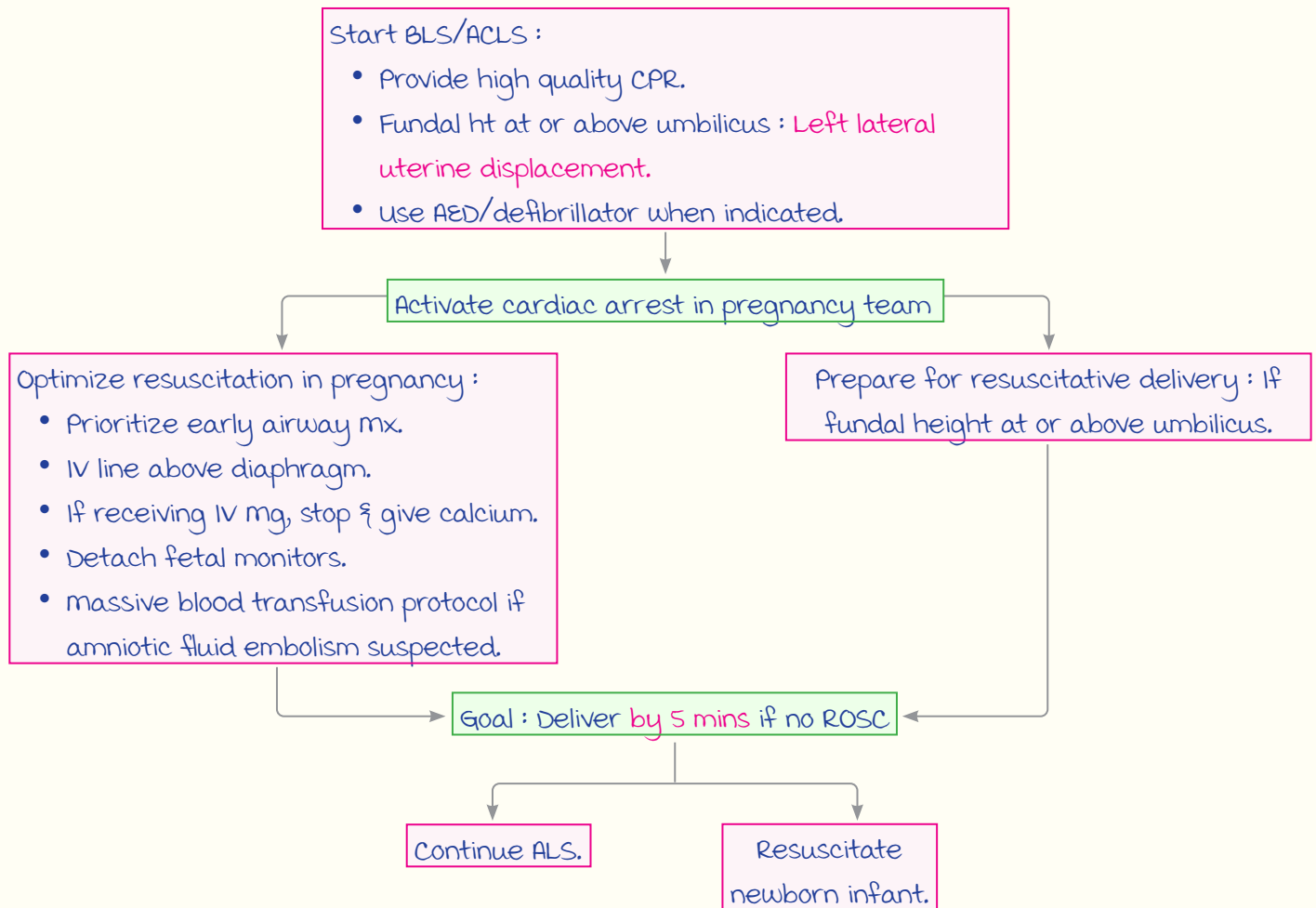
00:45:30



Cardiac Arrest in Pregnancy Algorithm

00:46:46

----- Active space -----



Etiology :

- **A**naesthesia complication.
- **B**leeding.
- **C**ardiovascular.
- **D**rugs.
- **E**mbolic (Amniotic fluid/Pulmonary).
- **F**ever
- **G**eneral causes (Hs & Ts).
- **H**ypertension.

